

Linear model 3



02/02/2026

Plan for today

- Quick recap
- Multiple regression
 - two continuous predictors
 - one categorical predictor
 - one continuous and one categorical predictor
- Interactions
- `lm()` output

Quick recap

Quick recap: Regression (conceptual)

Linear model: Simple regression

Data = Model + Error

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$\hat{Y}_i = \beta_0 + \beta_1 X_i$$

the model is a linear combination of predictors

The general procedure

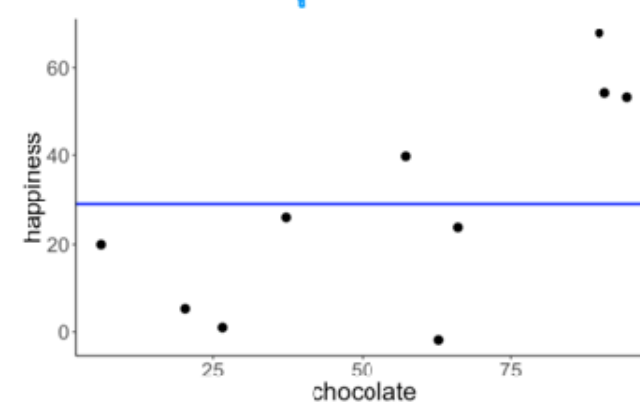
1. Define H_0 as Model C (compact) and H_1 as Model A (augmented)
2. Fit model parameters to the data
3. Calculate the proportional reduction of error (PRE) in our sample
4. Decide whether the augmented model is **worth it** by comparing the observed PRE in our sample to the sampling distribution of PRE (assuming that H_0 is true)

H_0 : Chocolate consumption and happiness are unrelated.

Model C

$$Y_i = \beta_0 + \epsilon_i$$

Model prediction



Fitted model

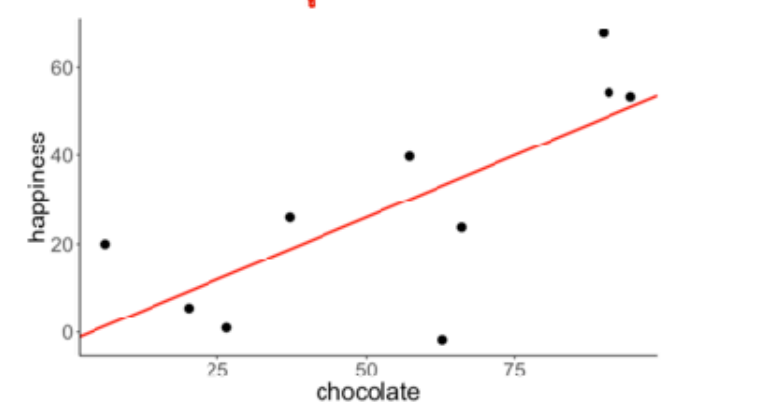
$$Y_i = 28.88 + e_i$$

H_1 : Chocolate consumption and happiness are related.

Model A

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Model prediction



Fitted model

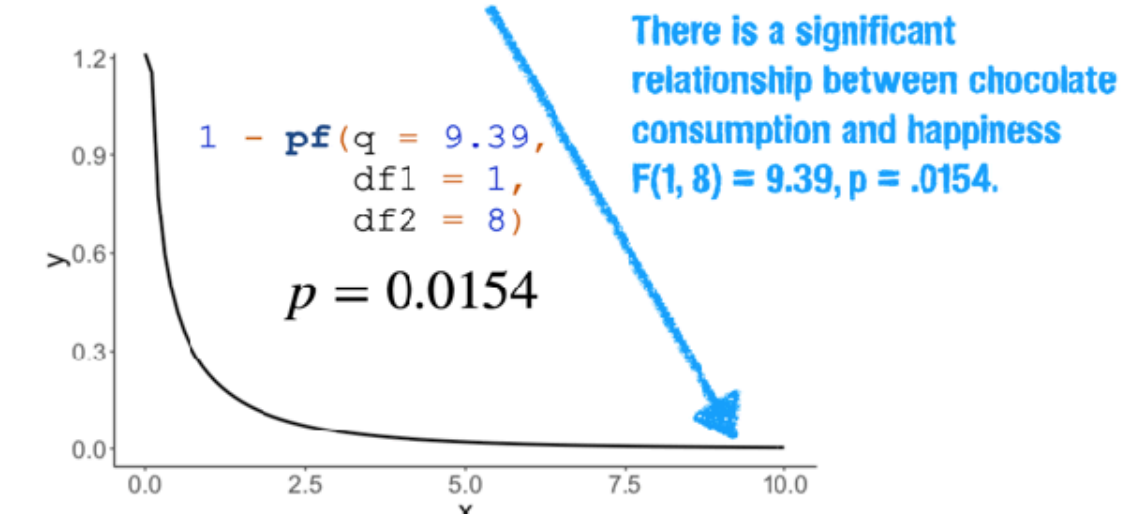
$$Y_i = -2.04 + 0.56X_i + e_i$$

Decide whether it's **worth it**

- To compute the F statistic, we need:

- PRE = 0.54
- PC = 1
- PA = 2
- $n = 10$

$$F = \frac{\text{PRE}/(\text{PA} - \text{PC})}{(1 - \text{PRE})/(n - \text{PA})}$$
$$= \frac{0.54/(2 - 1)}{(1 - 0.54)/(10 - 2)}$$
$$= 9.39$$



Quick recap: Regression (in R)

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$\text{balance}_i = b_0 + b_1 \cdot \text{income}_i + e_i$$

```
fit = lm(formula = balance ~ 1 + income, data = df.credit)
```

outcome intercept predictor data

(doesn't need to be specified explicitly)

```
anova(fit_c, fit_a)
```

```
Analysis of Variance Table

Model 1: balance ~ 1
Model 2: balance ~ 1 + income
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1     399 84339912
2     398 66208745  1  18131167 108.99 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\text{PRE} = 1 - \frac{66208745}{84339912} \approx 0.215$$

The augmented model reduces the error by 21.5%.

```
lm(balance ~ 1 + income, data = df.credit) %>%
  summary()
```

R^2

```
Residual standard error: 407.9 on 398 degrees of freedom
Multiple R-squared: 0.215, Adjusted R-squared: 0.213
F-statistic: 109 on 1 and 398 DF, p-value: < 2.2e-16
```



The general procedure

1. Define H_0 as Model C (compact) and H_1 as Model A (augmented)
2. Fit model parameters to the data

```
fit_c = lm(formula = balance ~ 1,
            data = df.credit)
```

```
fit_a = lm(formula = balance ~ 1 + income,
            data = df.credit)
```

3. Calculate the proportional reduction of error (PRE) in our sample

4. Decide whether the augmented model is **worth it** by comparing the observed PRE in our sample to the sampling distribution of PRE (assuming that H_0 is true)

```
anova(fit_c, fit_a)
```

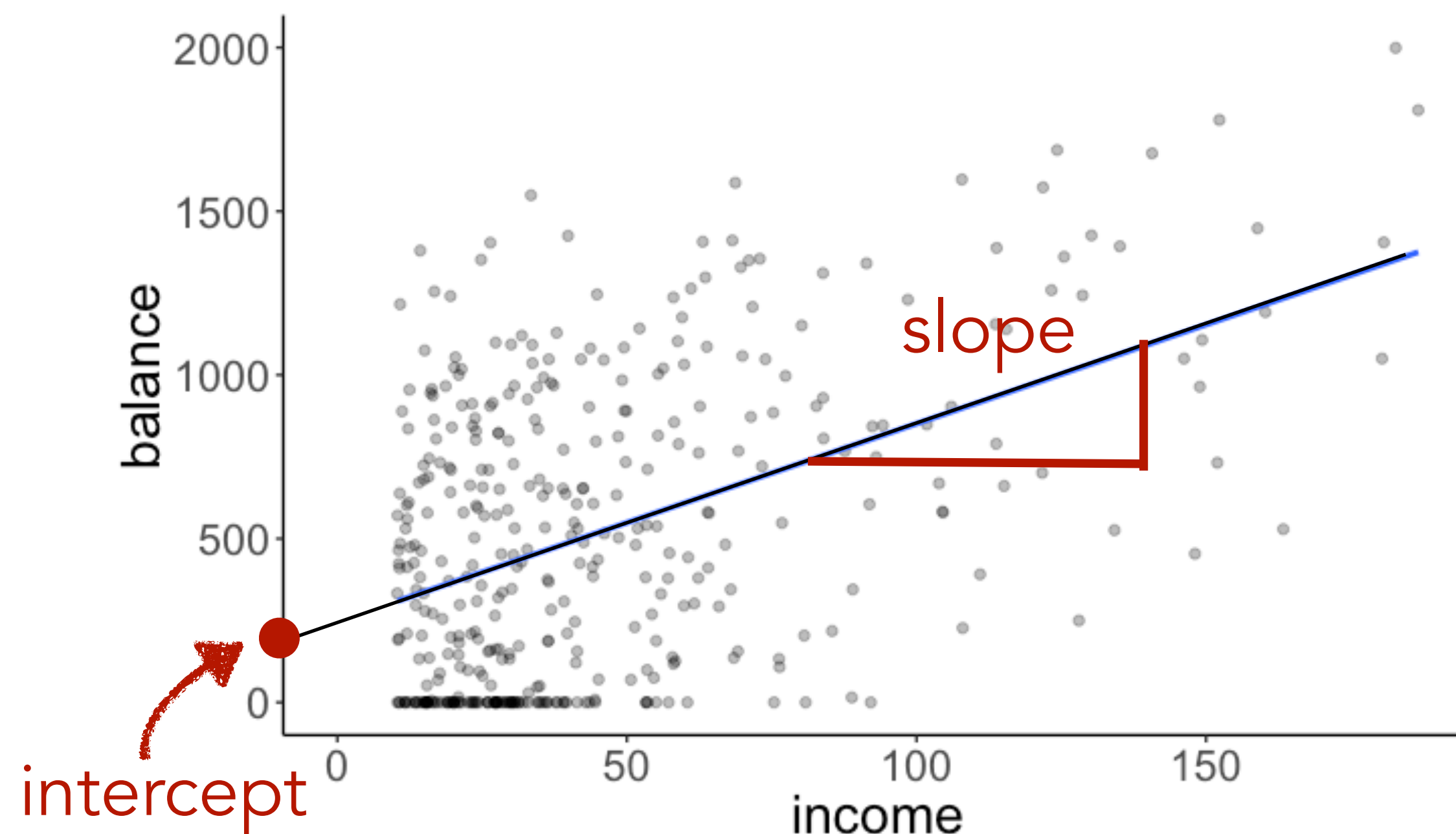
Interpreting regression parameters

Coefficients:	
(Intercept)	income
246.515	6.048

variable	description
income	in thousand dollars
balance	average credit card debt in dollars

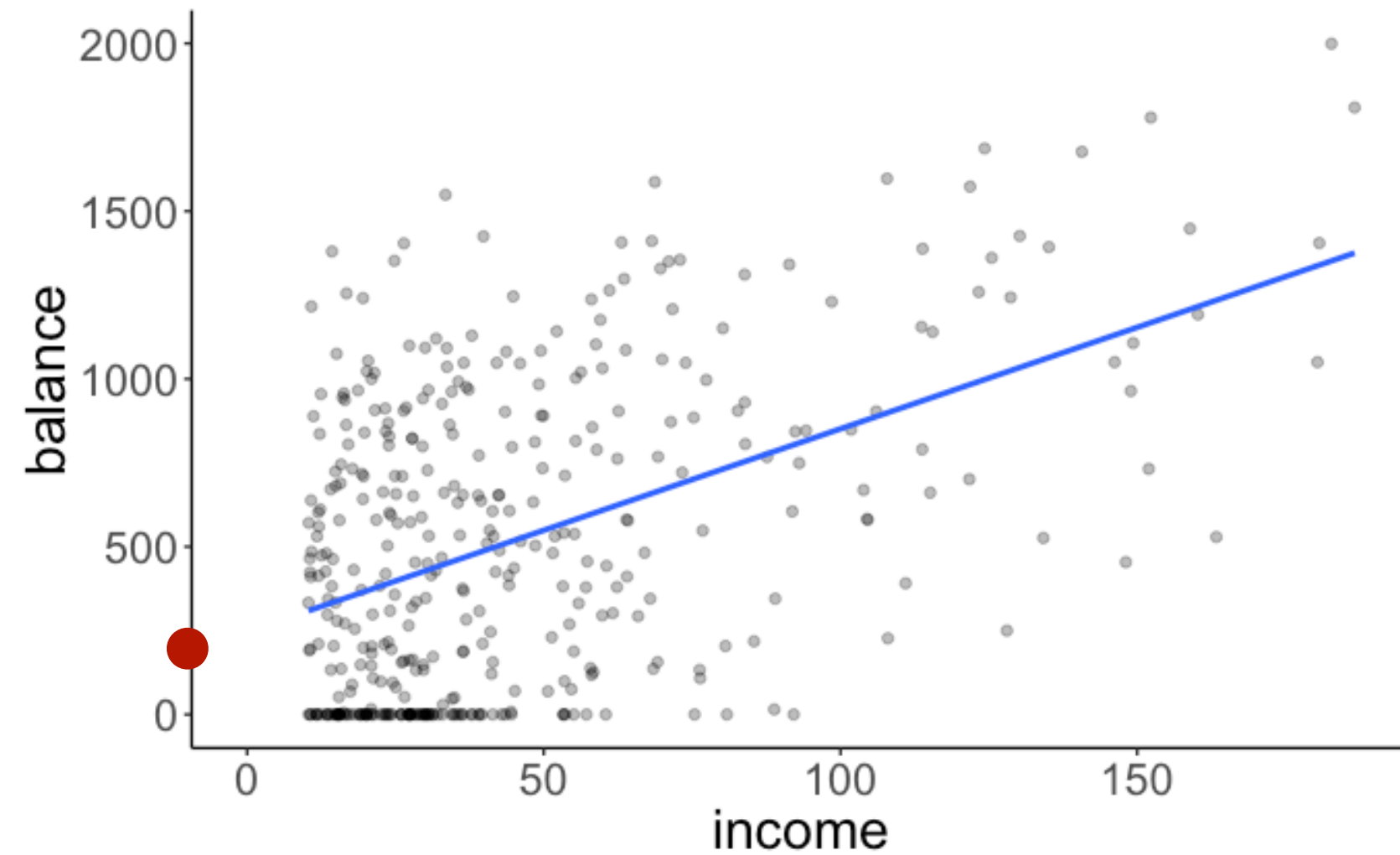
$$\text{balance}_i = b_0 + b_1 \cdot \text{income}_i + e_i$$

$$\text{balance}_i = 246.515 \cdot 1 + 6.048 \cdot \text{income}_i + e_i$$

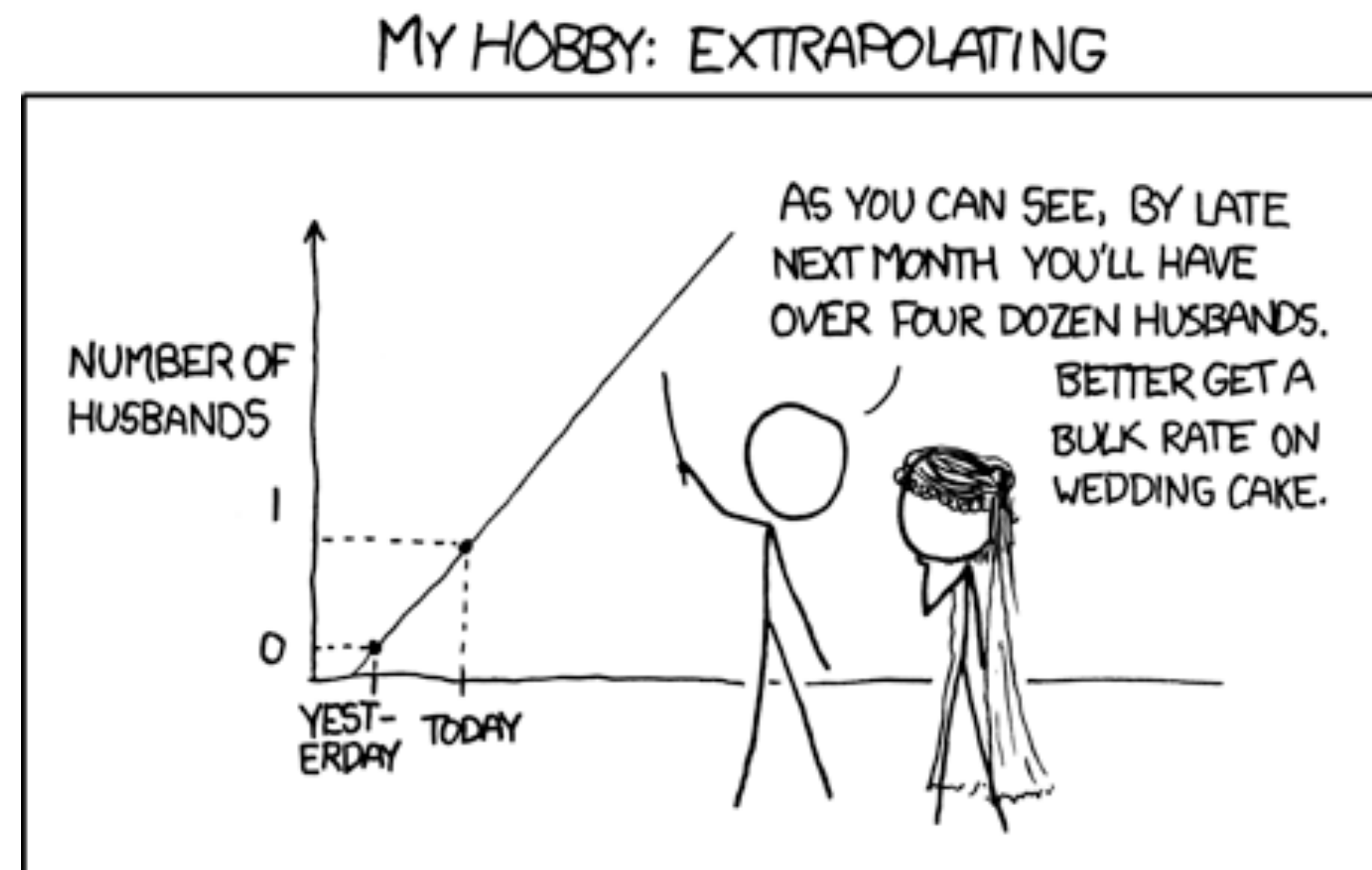


For each additional thousand dollars of income, people carry average credit card balance that is \$6.05 higher.

Be careful about extrapolating predictions



- intercept is often outside the range of predictor values
- sometimes doesn't make sense (e.g. age = 0, height = 0, ...)



summary ()

```
1 lm(balance ~ 1 + income, data = df.credit) %>%  
2   summary()
```

```
Call:  
lm(formula = balance ~ 1 + income, data = df.credit)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-803.64 -348.99  -54.42   331.75 1100.25   
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept)  246.5148     33.1993   7.425  6.9e-13 ***  
income        6.0484      0.5794  10.440  < 2e-16 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 407.9 on 398 degrees of freedom  
Multiple R-squared:  0.215, Adjusted R-squared:  0.213  
F-statistic: 109 on 1 and 398 DF, p-value: < 2.2e-16
```




```
library("broom")
```

helps with tidying up
model objects in R

augment () adds columns to the original data such as predictions, residuals
and cluster assignments

tidy () summarizes a model's statistical findings such as coefficients of a
regression

glance () provides a one-row summary of model-level statistics

broom: turn messy model outputs
into tidy **TIBBLES!**



@allison-horst

augment ()

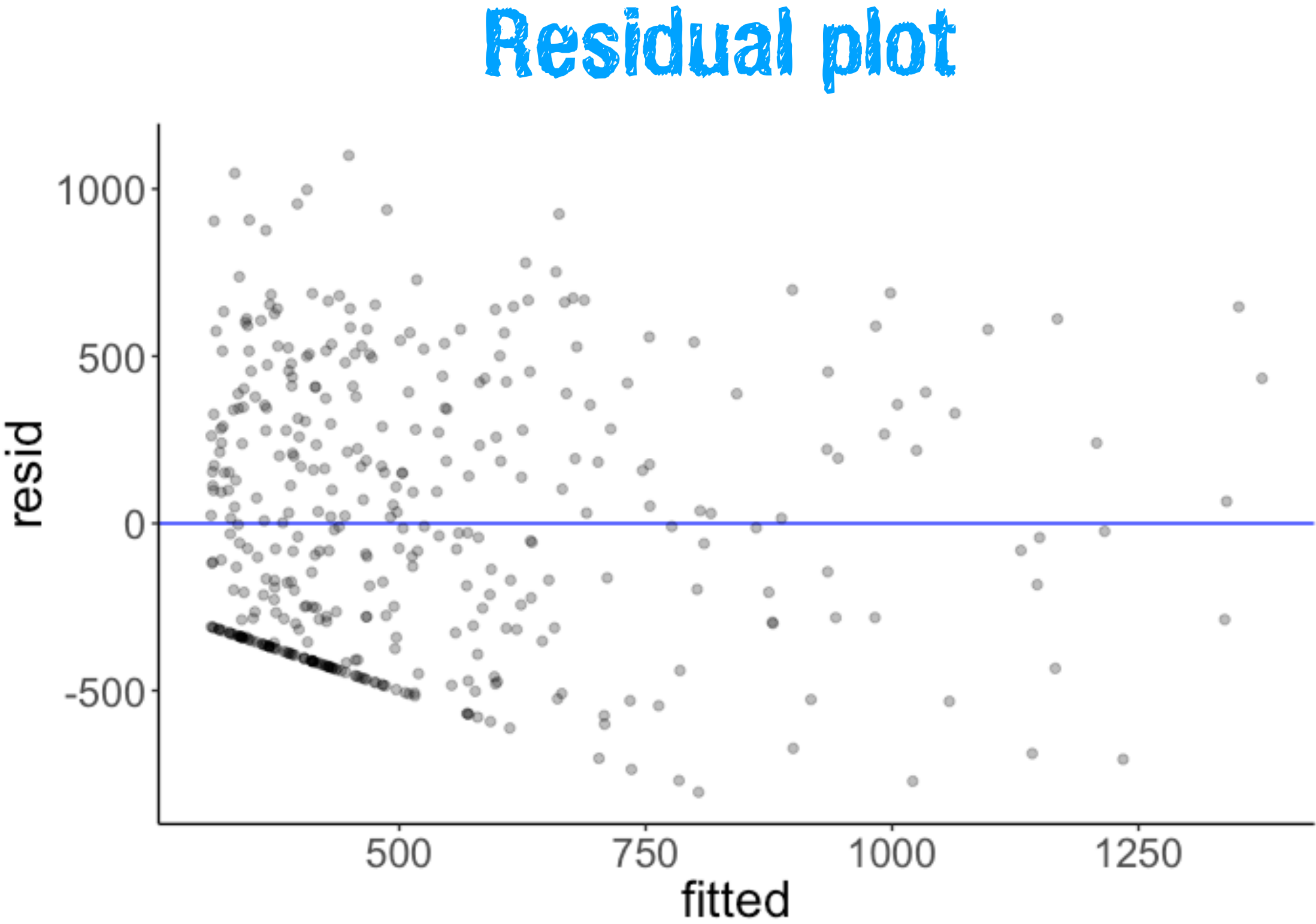
fit %>%
augment ()

balance	income	.fitted	.se.fit	.resid	.hat	.sigma	.cooksd	.std.resid
333	14.89	336.58	26.92	-3.58	0.00	408.38	0.00	-0.01
903	106.03	887.79	40.71	15.21	0.01	408.38	0.00	0.04
580	104.59	879.13	39.99	-299.13	0.01	408.10	0.00	-0.74
964	148.92	1147.26	63.45	-183.26	0.02	408.27	0.00	-0.45
331	55.88	584.51	21.31	-253.51	0.00	408.18	0.00	-0.62
1151	80.18	731.47	28.74	419.53	0.00	407.83	0.00	1.03
203	21.00	373.51	24.76	-170.51	0.00	408.29	0.00	-0.42
872	71.41	678.42	25.42	193.58	0.00	408.26	0.00	0.48
279	15.12	338.00	26.83	-59.00	0.00	408.37	0.00	-0.14
1350	71.06	676.32	25.30	673.68	0.00	406.97	0.01	1.65

augment ()

```
fit %>%  
  augment ( )
```

balance	income	.fitted	.resid
333	14.89	336.58	-3.58
903	106.03	887.79	15.21
580	104.59	879.13	-299.13
964	148.92	1147.26	-183.26
331	55.88	584.51	-253.51
1151	80.18	731.47	419.53
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872	71.41	678.42	193.58
279	15.12	338.00	-59.00
1350	71.06	676.32	673.68

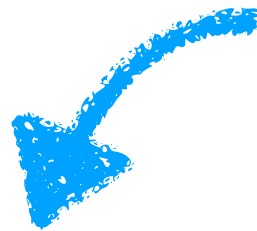


tidy()

```
1 lm(balance ~ 1 + income, data = df.credit) %>%  
2   summary()
```

```
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)  
(Intercept) 246.5148    33.1993   7.425  6.9e-13 ***  
income       6.0484     0.5794  10.440  < 2e-16 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
1 fit %>%  
2   tidy(conf.int = TRUE)
```

 a data frame, yay!

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	246.51	33.20	7.43	0	181.25	311.78
income	6.05	0.58	10.44	0	4.91	7.19

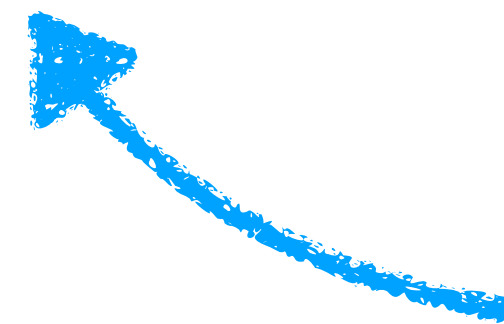
glance ()

```
1 lm(balance ~ 1 + income, data = df.credit) %>%  
2   summary()
```

```
Residual standard error: 407.9 on 398 degrees of freedom  
Multiple R-squared:  0.215,    Adjusted R-squared:  0.213  
F-statistic:    109 on 1 and 398 DF,  p-value: < 2.2e-16
```

```
1 fit %>%  
2   glance()
```

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.21	0.21	407.86	108.99	0	2	-2970.95	5947.89	5959.87	66208745	398



useful model summary
(we will learn later what
the different values mean)

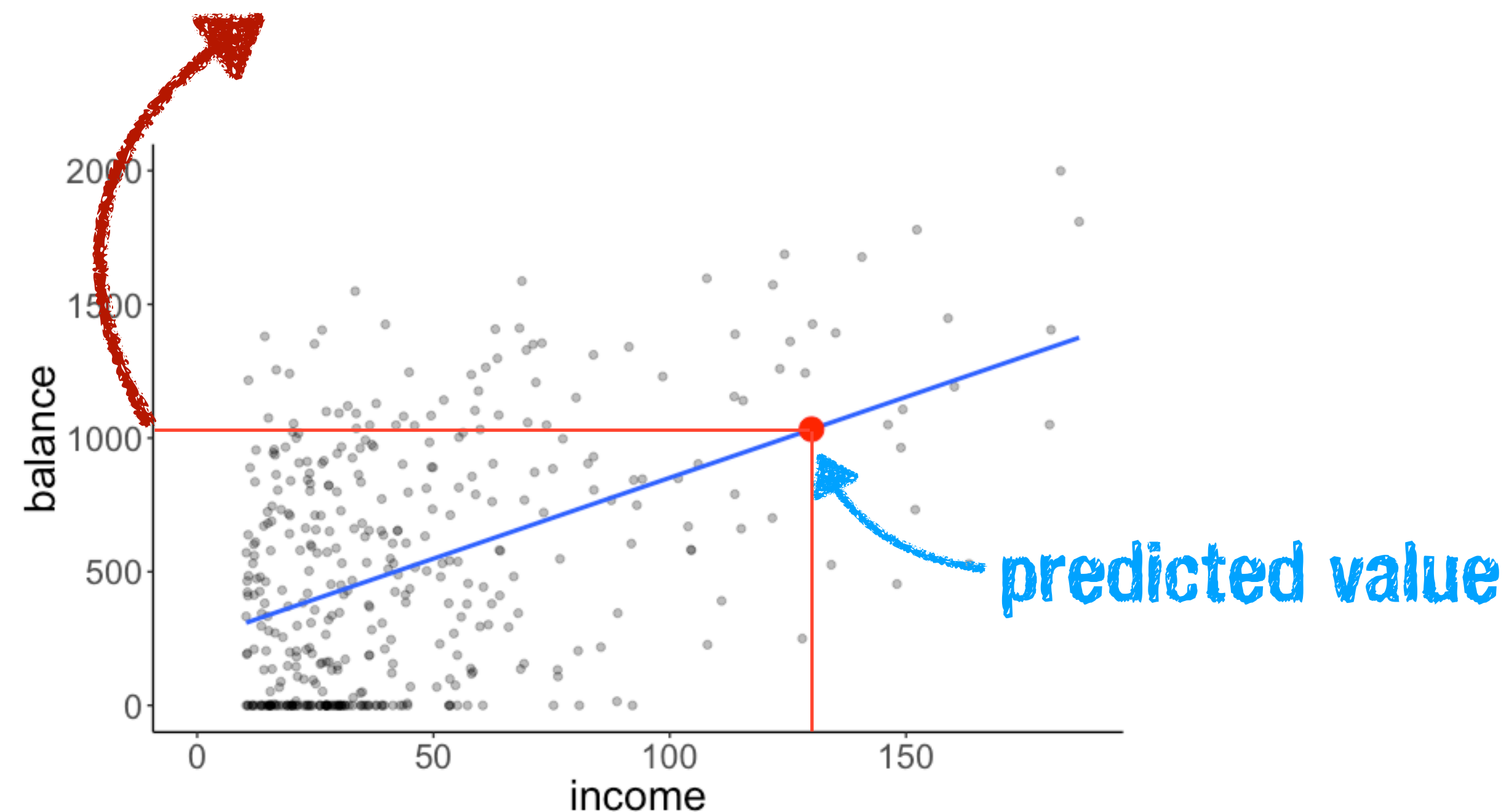
Making predictions

```
fit = lm(balance ~ 1 + income, data = df.credit)
```

$$\text{balance}_i = 246.515 + 6.048 \cdot \text{income}_i + e_i$$

```
augment(fit, newdata = tibble(income = 130))
```

$$\widehat{\text{balance}} = 246.515 + 6.048 \cdot 130$$



Hypothesis test

Compact Model

$$\text{balance}_i = \beta_0 + \epsilon_i$$

```
fit_c = lm(formula = balance ~ 1,  
            data = df.credit)
```

Augmented Model

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i + \epsilon_i$$

```
fit_a = lm(formula = balance ~ 1 + income,  
            data = df.credit)
```

```
anova(fit_c, fit_a)
```

```
Analysis of Variance Table

Model 1: balance ~ 1
Model 2: balance ~ 1 + income
   Res.Df    RSS Df Sum of Sq   F    Pr(>F)
1     399 84339912
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The general procedure

1. Define H_0 as Model C (compact) and H_1 as Model A (augmented)

```
fit_c = lm(formula = balance ~ 1,  
           data = df.credit)
```

2. Fit model parameters to the data

```
fit_a = lm(formula = balance ~ 1 + income,  
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```

3. Calculate the proportional reduction of error (PRE) in our sample

4. Decide whether the augmented model is **worth it** by comparing the observed PRE in our sample to the sampling distribution of PRE (assuming that H_0 is true)

```
anova(fit_c, fit_a)
```


Hypothesis test

```
anova (fit_c, fit_a)
```

```
Analysis of Variance Table

Model 1: balance ~ 1
Model 2: balance ~ 1 + income
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
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---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\text{PRE} = 1 - \frac{66208745}{84339912} \approx 0.215$$

The augmented model reduces the error by 21.5%.

```
lm(balance ~ 1 + income, data = df.credit) %>%
summary()
```

R^2

```
Residual standard error: 407.9 on 398 degrees of freedom
Multiple R-squared: 0.215, Adjusted R-squared: 0.213
F-statistic: 109 on 1 and 398 DF, p-value: < 2.2e-16
```

Hypothesis test

the **compact model** predicts the mean (which doesn't explain any of the variance)

- in the case of a simple regression, PRE (proportion of reduced error) of comparing the full model versus a model with just an intercept is identical to R^2 (variance explained)
- and R^2 is directly related to the correlation coefficient r

```
cor(df.credit$balance,  
    df.credit$income)
```

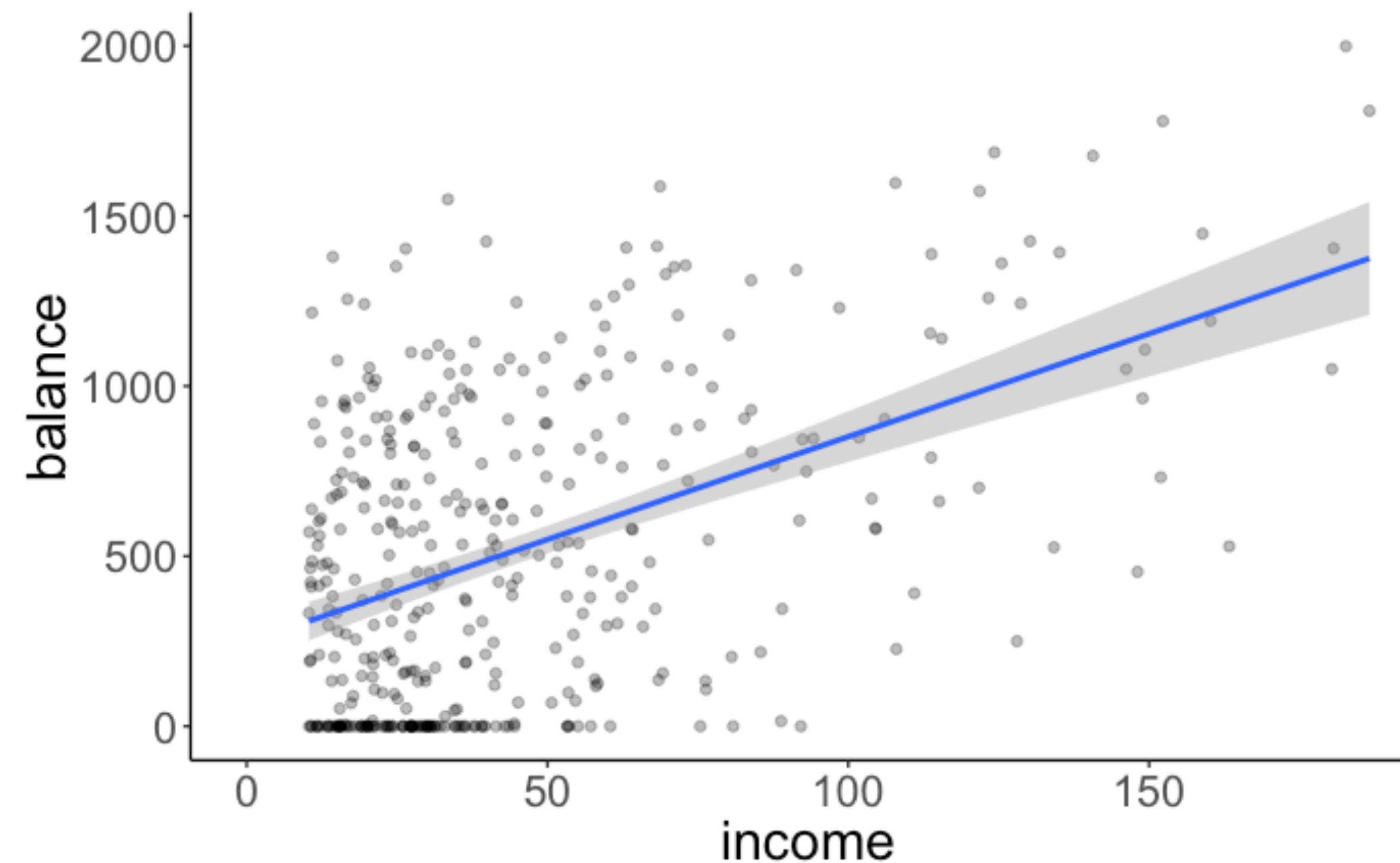
$$R^2 = 0.215$$

$$r = .463$$

effect size measure

Reporting results

plot



statistical

test There is a significant relationship between credit card holders' income and the average balance on their credit cards

effect size
measure

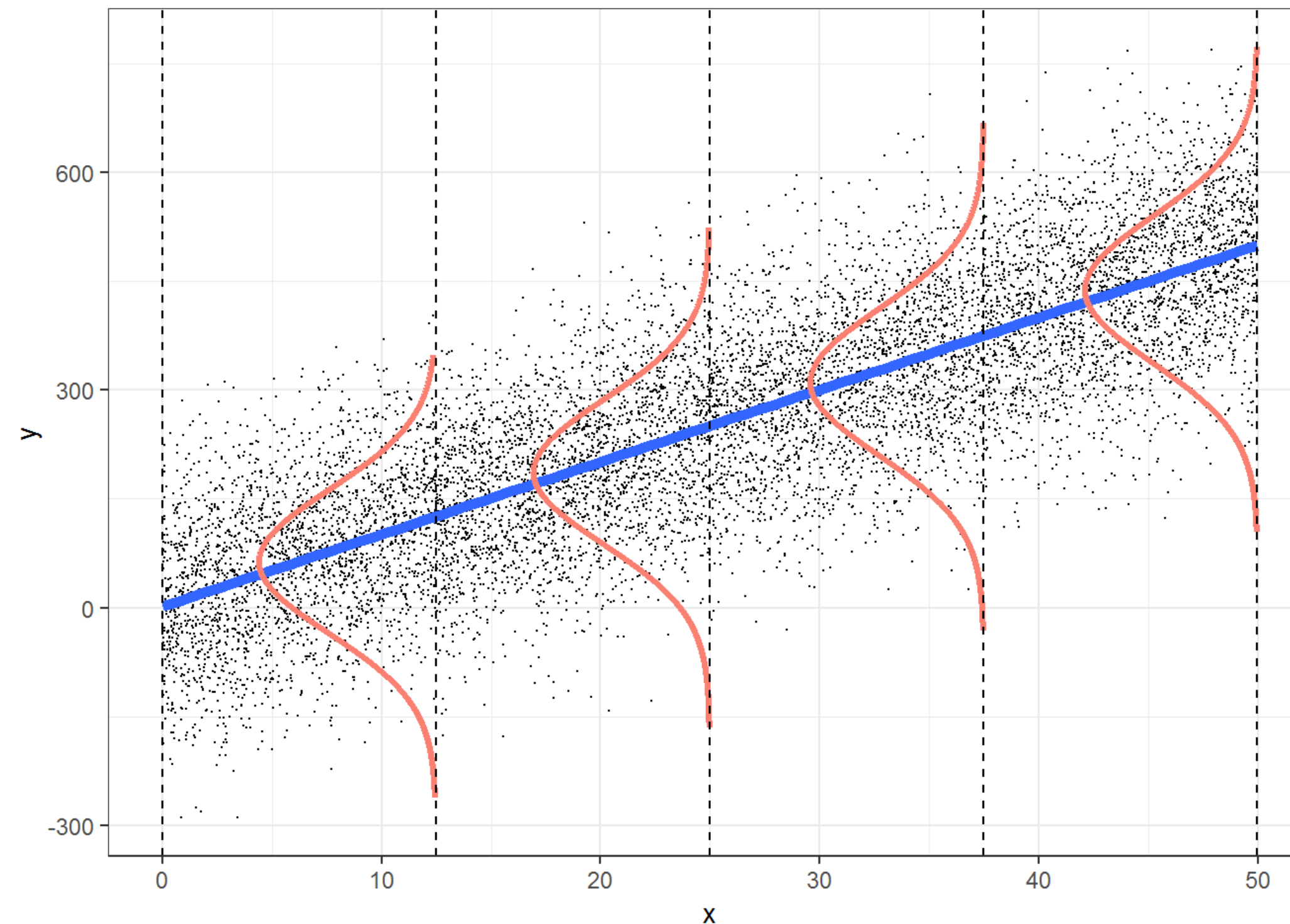
$F(1, 389) = 108.99, p < .001, r = .463.$

With each additional \$1000 of income, the average balance is predicted to be higher by \$6.05 [4.91, 7.19] (95% CI).

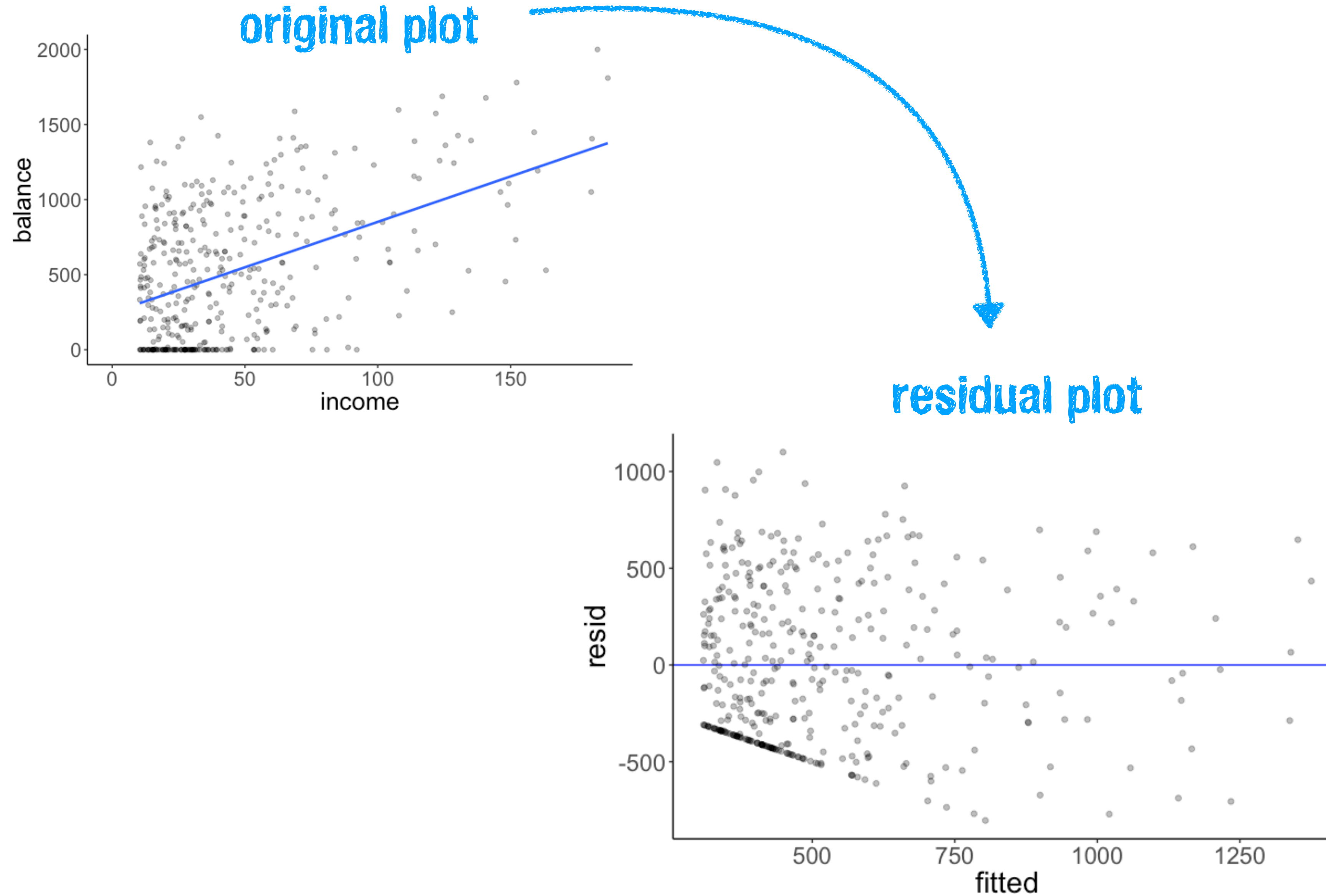
parameter estimate with confidence interval

Model assumptions

- independent observations
- Y is continuous
- errors are normally distributed
- errors have constant variance
- error terms are uncorrelated

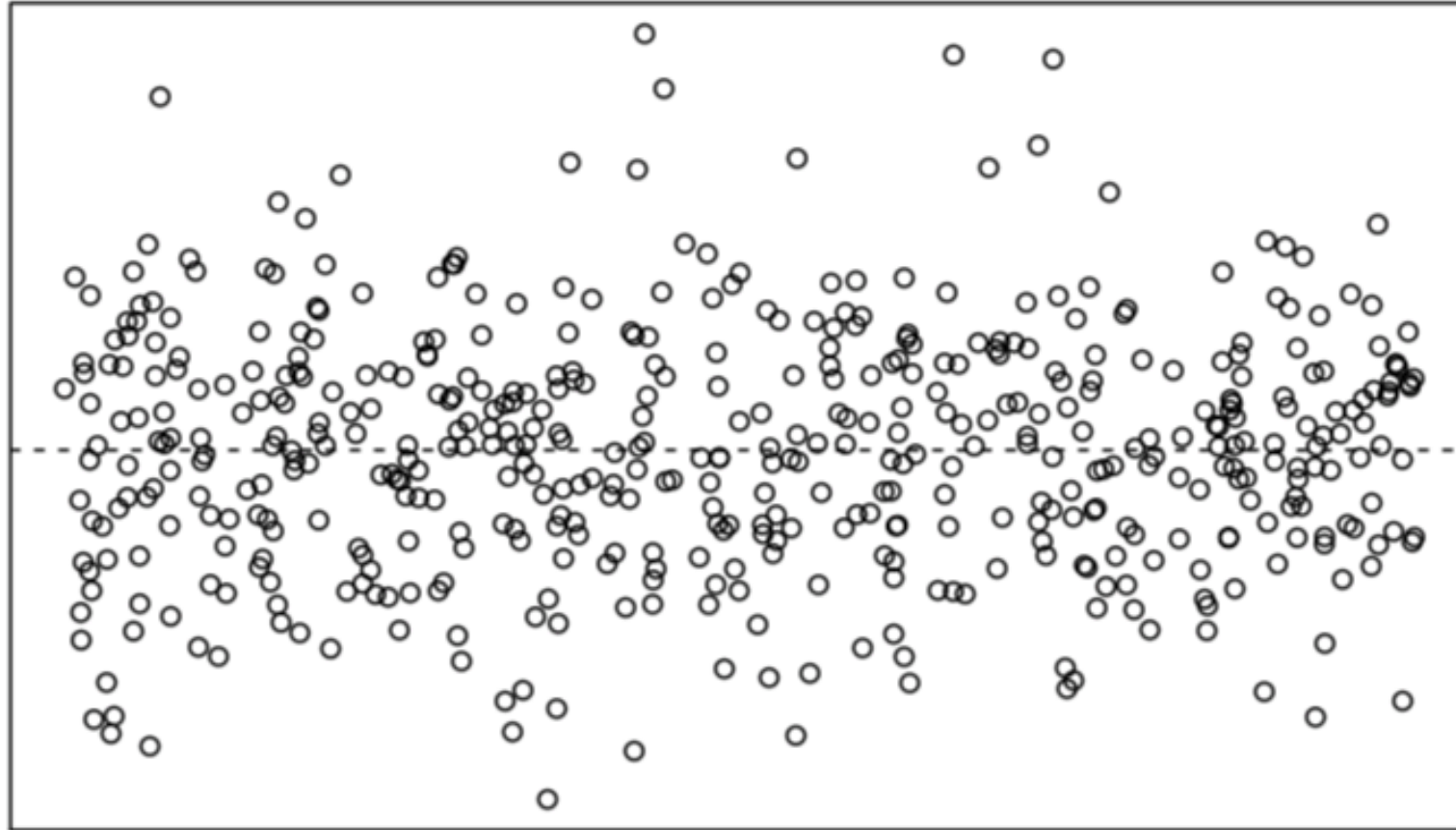


Model assumptions

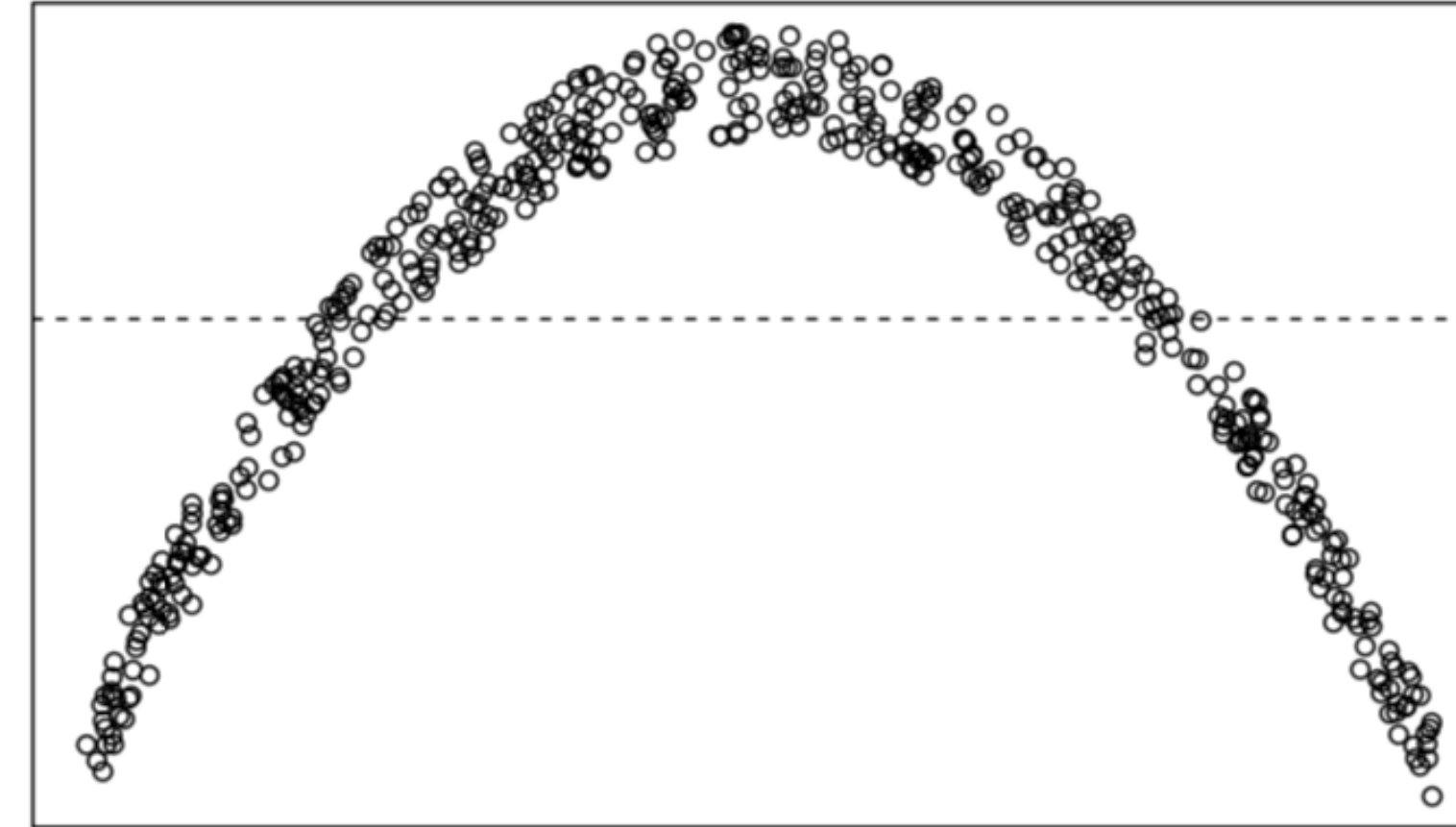


Model assumptions

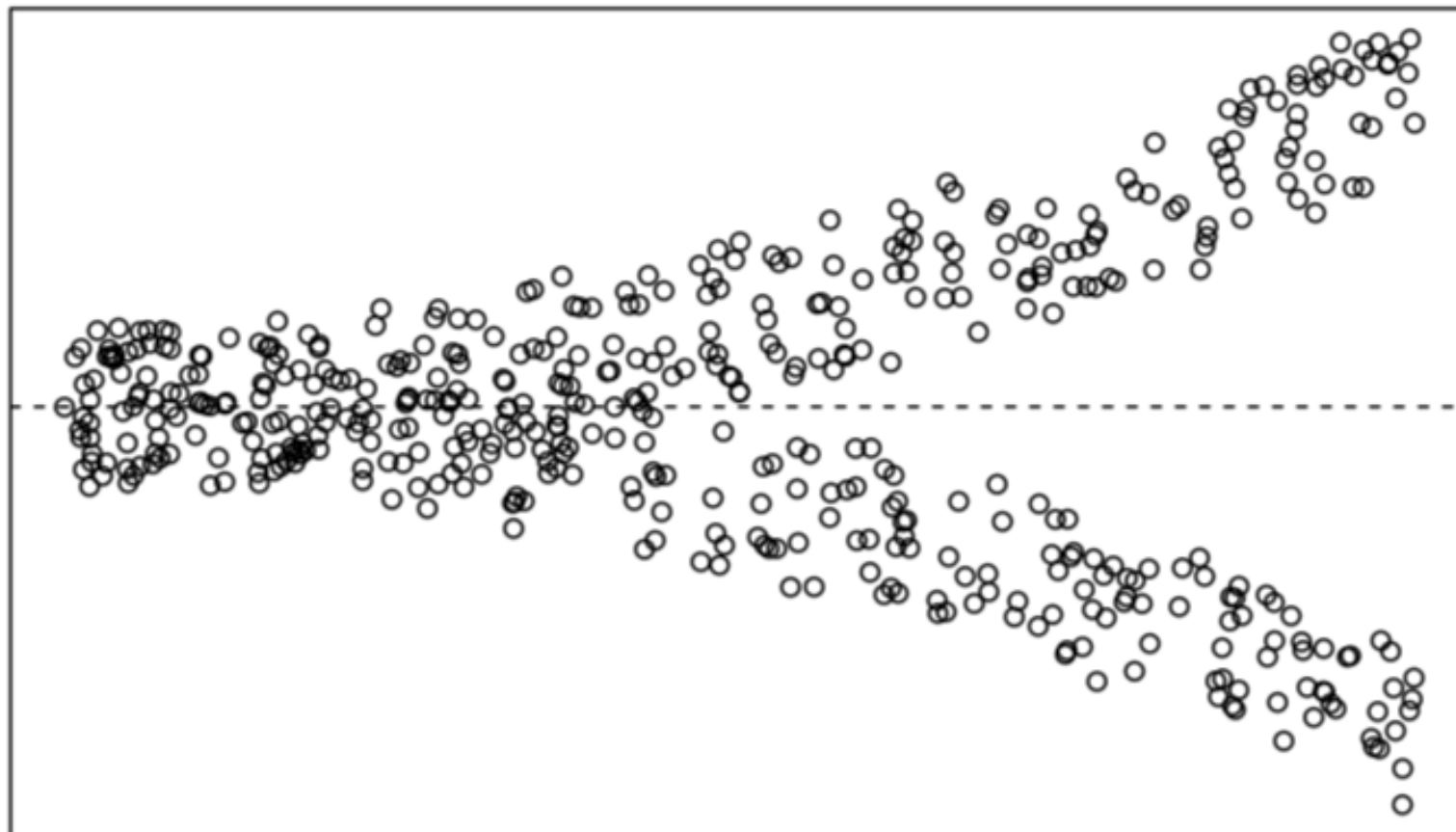
No Violation



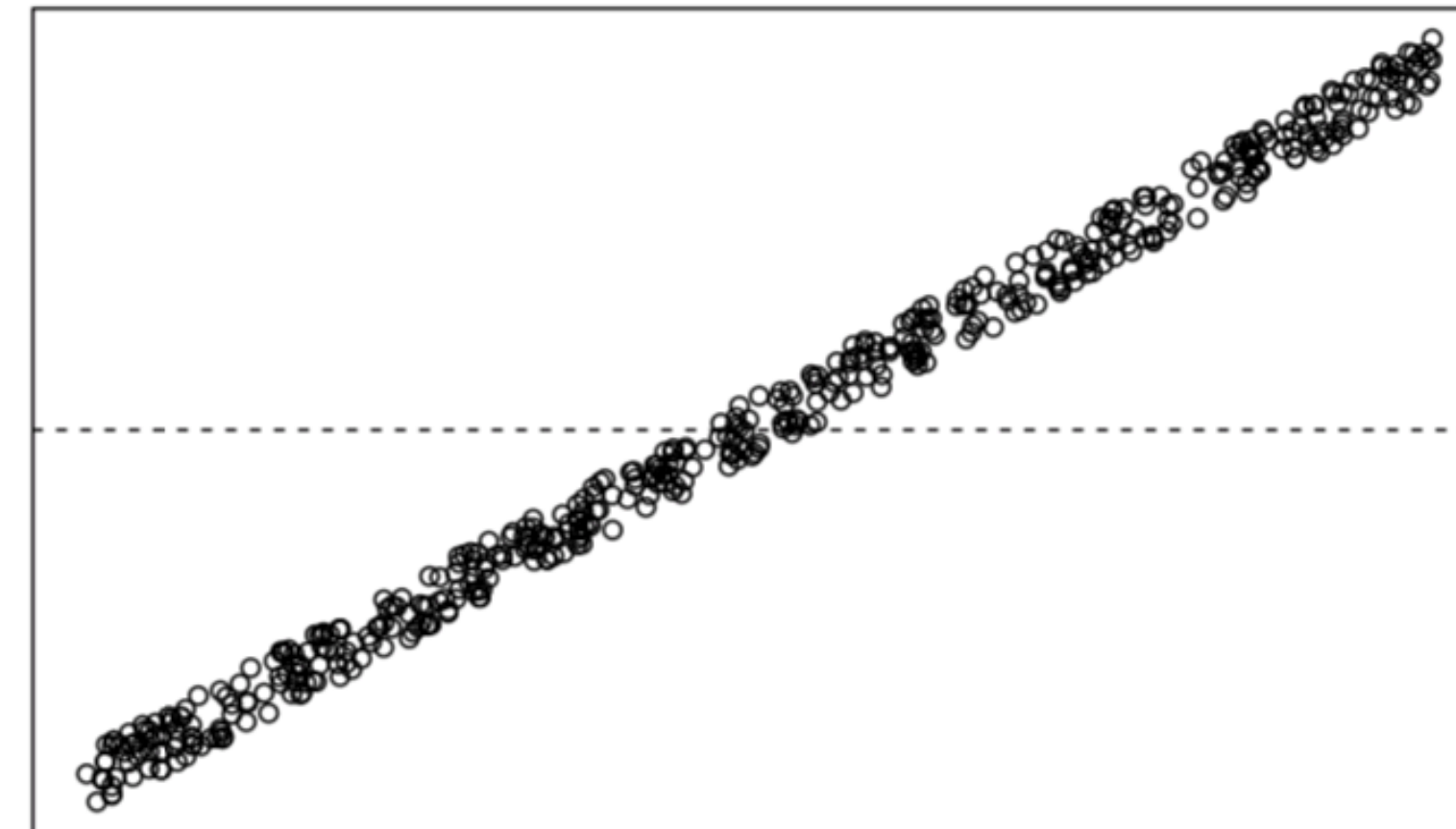
Nonlinear Relationship



Nonconstant Error Variance



Dependent Error Terms



Multiple regression

Linear model

$$\text{Data} = \text{Model} + \text{Error}$$

Simple regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i$$

one predictor



Multiple regression

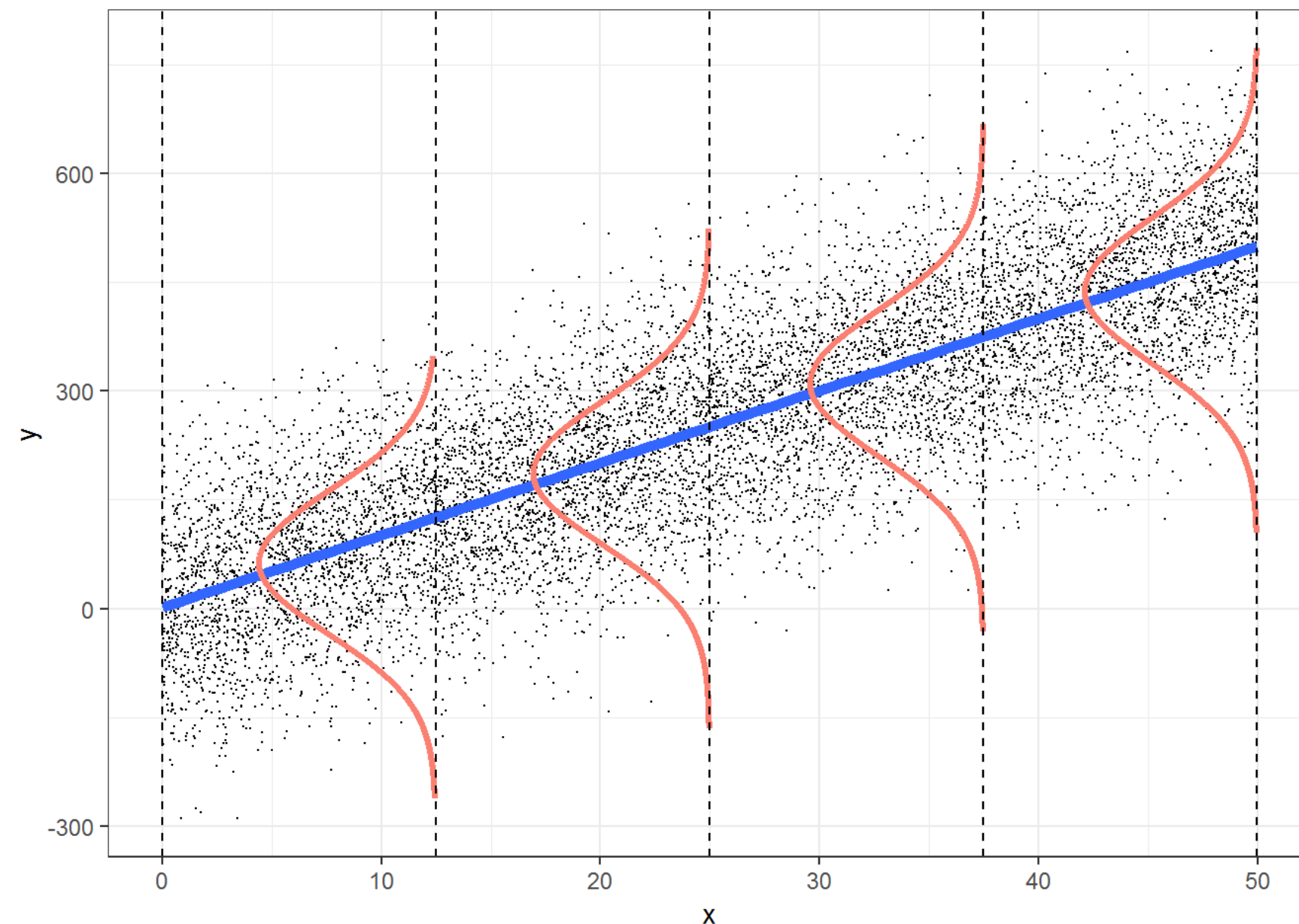
$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \epsilon_i$$

many predictors

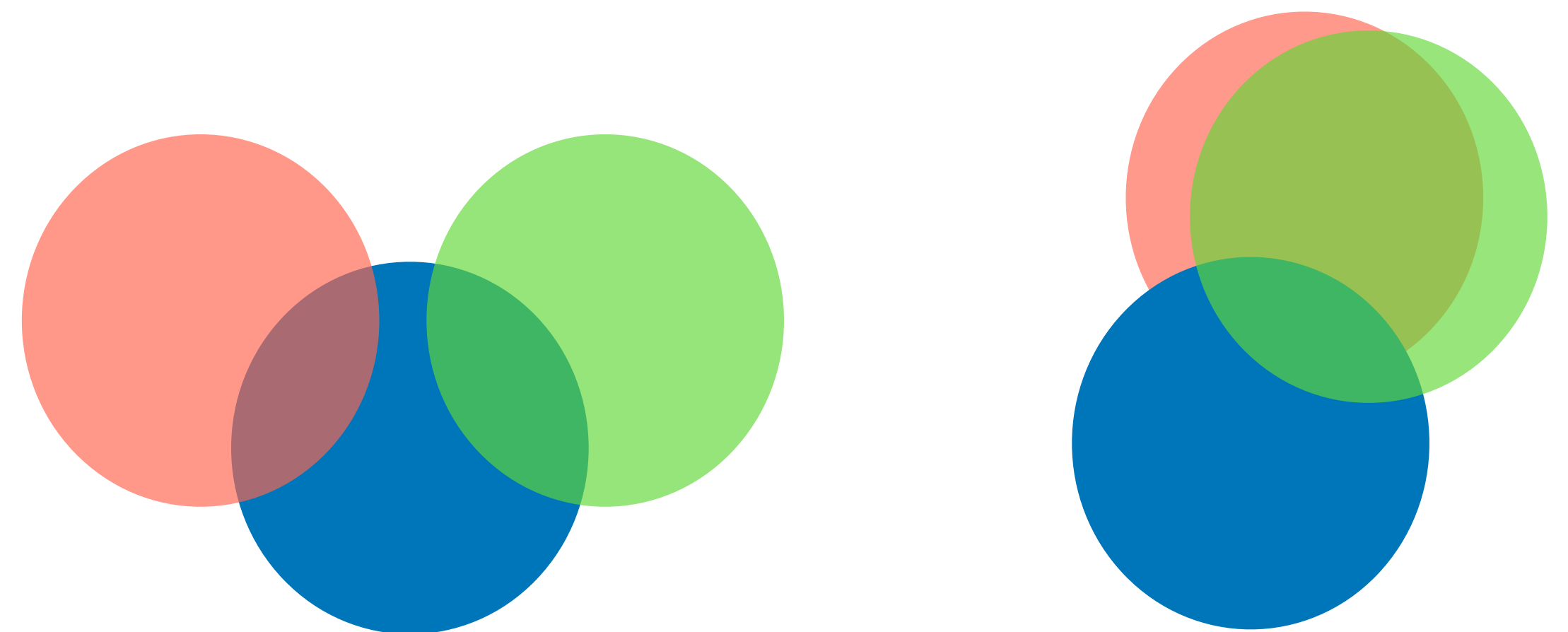


Assumptions of **multiple** regression

- independent observations
- Y is continuous
- errors are normally distributed
- errors have constant variance
- error terms are uncorrelated
- **no multicollinearity**



predictors in the model should not be highly correlated with each other



Advertising data set

money spent on
different media
(x \$1000)

sales
(x 1000)

- Combine several predictors to explain an outcome variable of interest

index	tv	radio	newspaper	sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75.0	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1.0	4.8
10	199.8	2.6	21.2	10.6

Model C

Simple regression

$$\text{sales}_i = b_0 + b_1 \cdot \text{tv}_i + e_i$$

Model A

Multiple regression

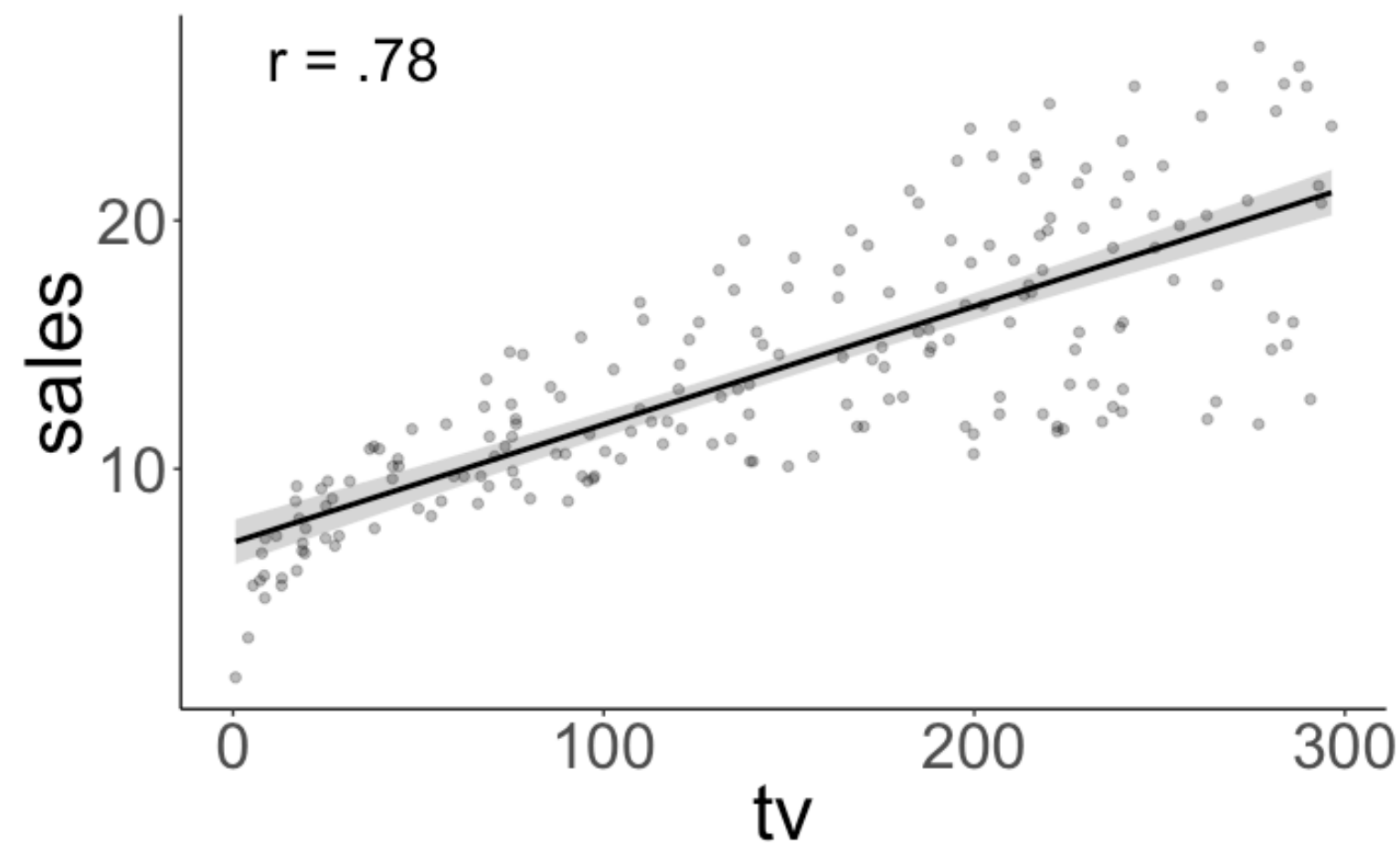
$$\text{sales}_i = b_0 + b_1 \cdot \text{tv}_i + b_2 \cdot \text{radio}_i + e_i$$

Can we predict sales better when we consider radio ads in addition to TV ads?

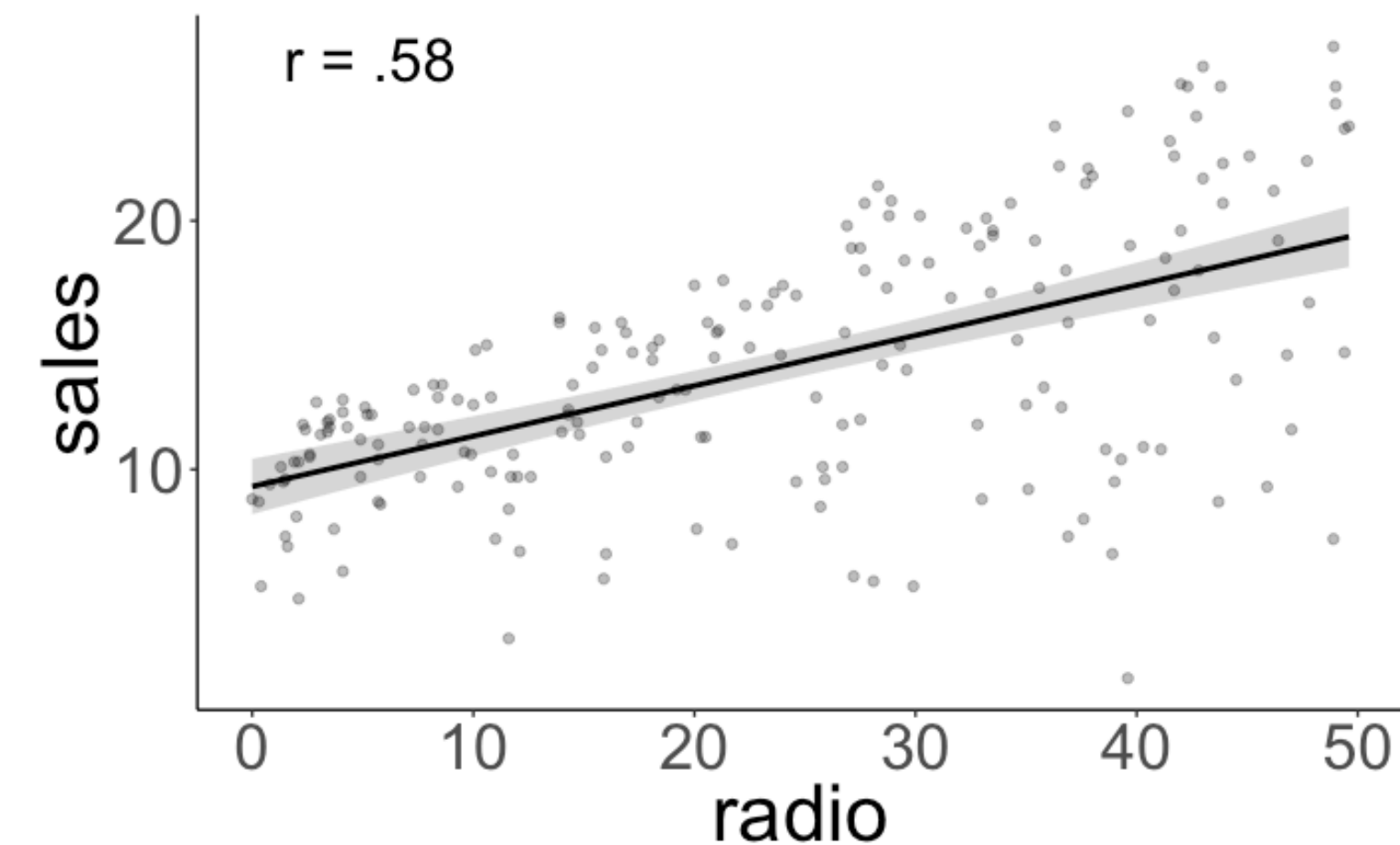
"Controlling" for TV ads, do radio ads explain any additional variance in sales?

$$\text{sales}_i = b_0 + b_1 \cdot \text{tv}_i + b_2 \cdot \text{radio}_i + e_i$$

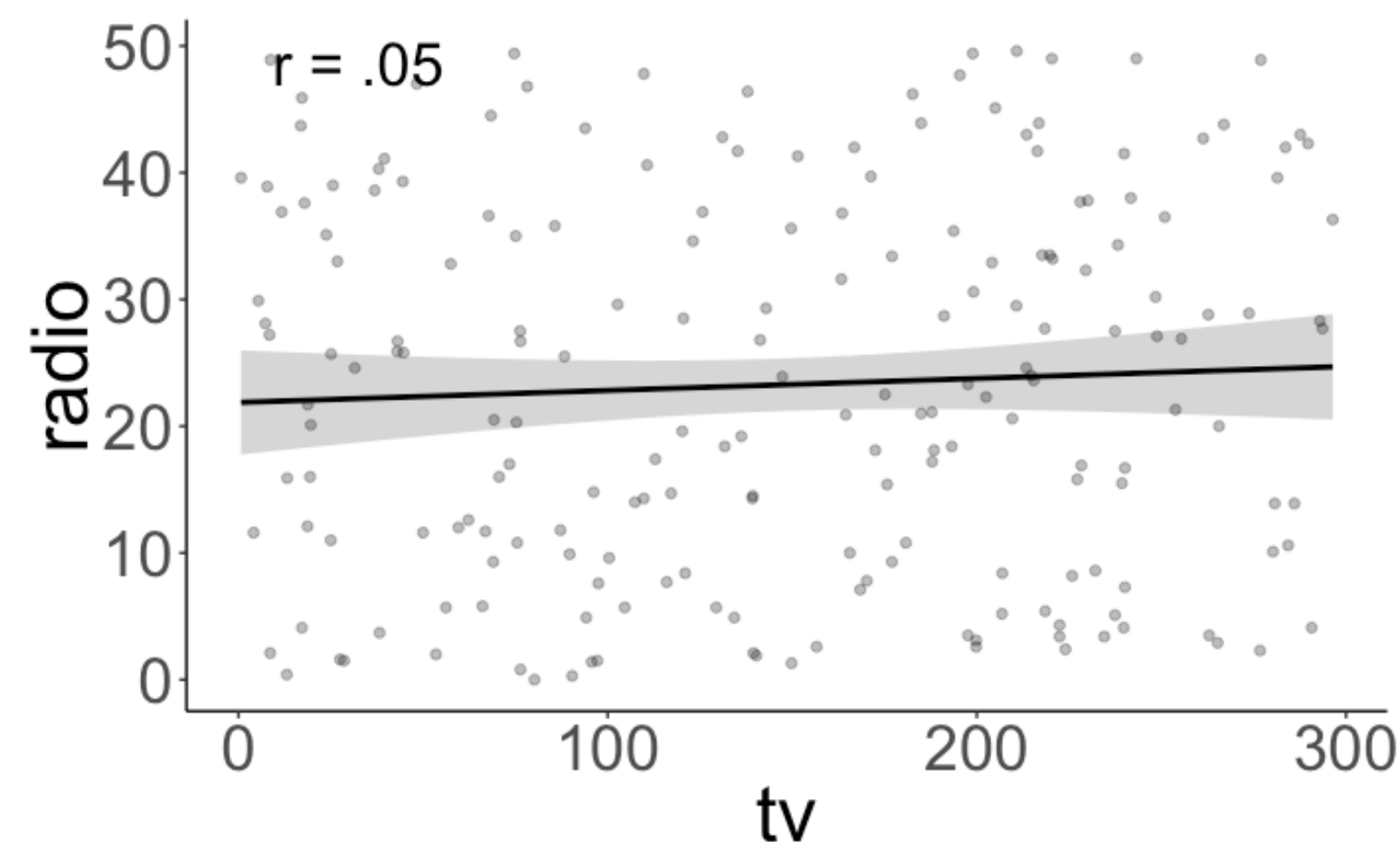
Relationship between TV ads and sales



Relationship between radio ads and sales



Relationship between TV ads and radio ads



**predictors are not
correlated, yay!**

Visualizing correlations

```
library("corrr")
```



```
1 df.credit %>%  
2   select_if(is.numeric) %>%  
3   correlate() %>%  
4   rearrange() %>%  
5   shave() %>%  
6   fashion()
```

rowname	index	newspaper	radio	sales	tv
index					
newspaper	-0.15				
radio	-0.11	0.35			
sales	-0.05	0.23	0.58		
tv	0.02	0.06	0.05	0.78	

Can we predict sales better when we consider radio in addition to TV ads?

1. Define H_0 as Model C (compact) and H_1 as Model A (augmented)
2. Fit model parameters to the data
3. Calculate the proportional reduction of error (PRE) in our sample
4. Decide whether the augmented model is **worth it** by comparing the observed PRE in our sample to the sampling distribution of PRE (assuming that H_0 is true)

H_0 : Radio ads and sales are not related once we control for TV ads.

Model C

$$\text{sales}_i = b_0 + b_1 \cdot \text{tv}_i + e_i$$

H_1 : Radio ads and sales are related even when we control for TV ads.

Model A

$$\text{sales}_i = b_0 + b_1 \cdot \text{tv}_i + b_2 \cdot \text{radio}_i + e_i$$

```
1 # fit the models
2 fit_c = lm(sales ~ 1 + tv, data = df.ads)
3 fit_a = lm(sales ~ 1 + tv + radio, data = df.ads)
4
5 # do the F test
6 anova(fit_c, fit_a)
```

we reject the H_0

Analysis of Variance Table

Model 1: sales ~ 1 + tv

Model 2: sales ~ 1 + tv + radio

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	198	2102.53				
2	197	556.91	1	1545.6	546.74	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Evaluating the model: Model fit

```
fit_a %>%  
  glance()
```

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.897	0.896	1.681	859.618	0	3	-386.197	780.394	793.587	556.914	197

r.squared	The percent of variance explained by the model
adj.r.squared	r.squared adjusted based on the degrees of freedom
sigma	The square root of the estimated residual variance
statistic	F-statistic
p.value	p-value from the F test, describing whether the full regression is significant
df	Degrees of freedom used by the coefficients
logLik	the data's log-likelihood under the model
AIC	the Akaike Information Criterion
BIC	the Bayesian Information Criterion
deviance	deviance
df.residual	residual degrees of freedom

Evaluating the model: Model fit

```
fit_a %>%  
  glance()
```

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.897	0.896	1.681	859.618	0	3	-386.197	780.394	793.587	556.914	197

Default Compact Model

$$\text{sales}_i = b_0 + e_i$$

The augmented model reduces 89.7% of the error compared to a default compact model that just predicts the mean.

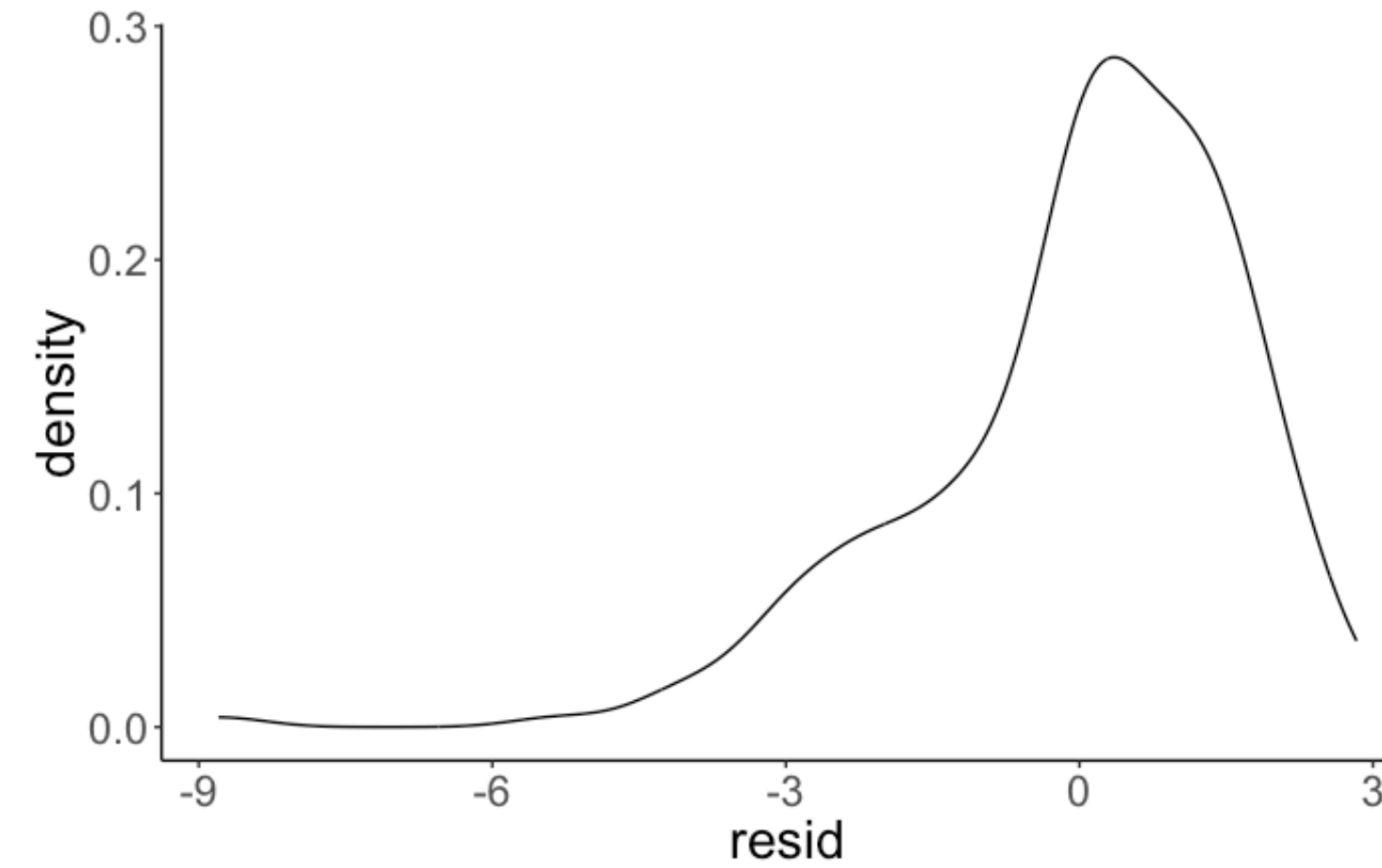
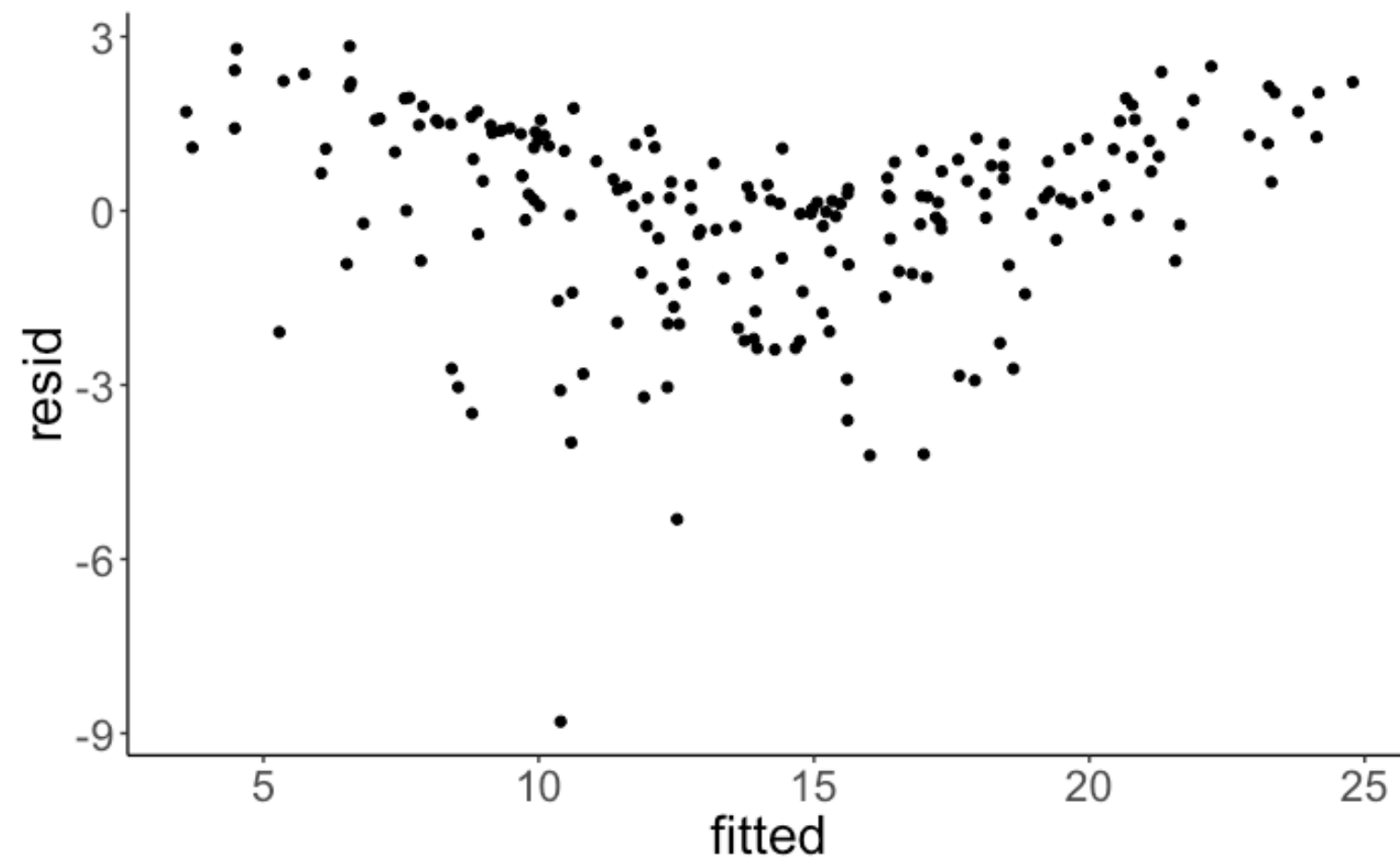
Augmented Model

$$\text{sales}_i = b_0 + b_1 \cdot \text{tv}_i + b_2 \cdot \text{radio}_i + e_i$$

$$\text{PRE} = 1 - \frac{\text{SSE}(A)}{\text{SSE}(C)} = R^2$$

Evaluating the model: Residual plots

```
resid = sales - fitted
```



OKish overall

Interpreting the results

```
fit_a %>%  
  tidy(conf.int = T)
```

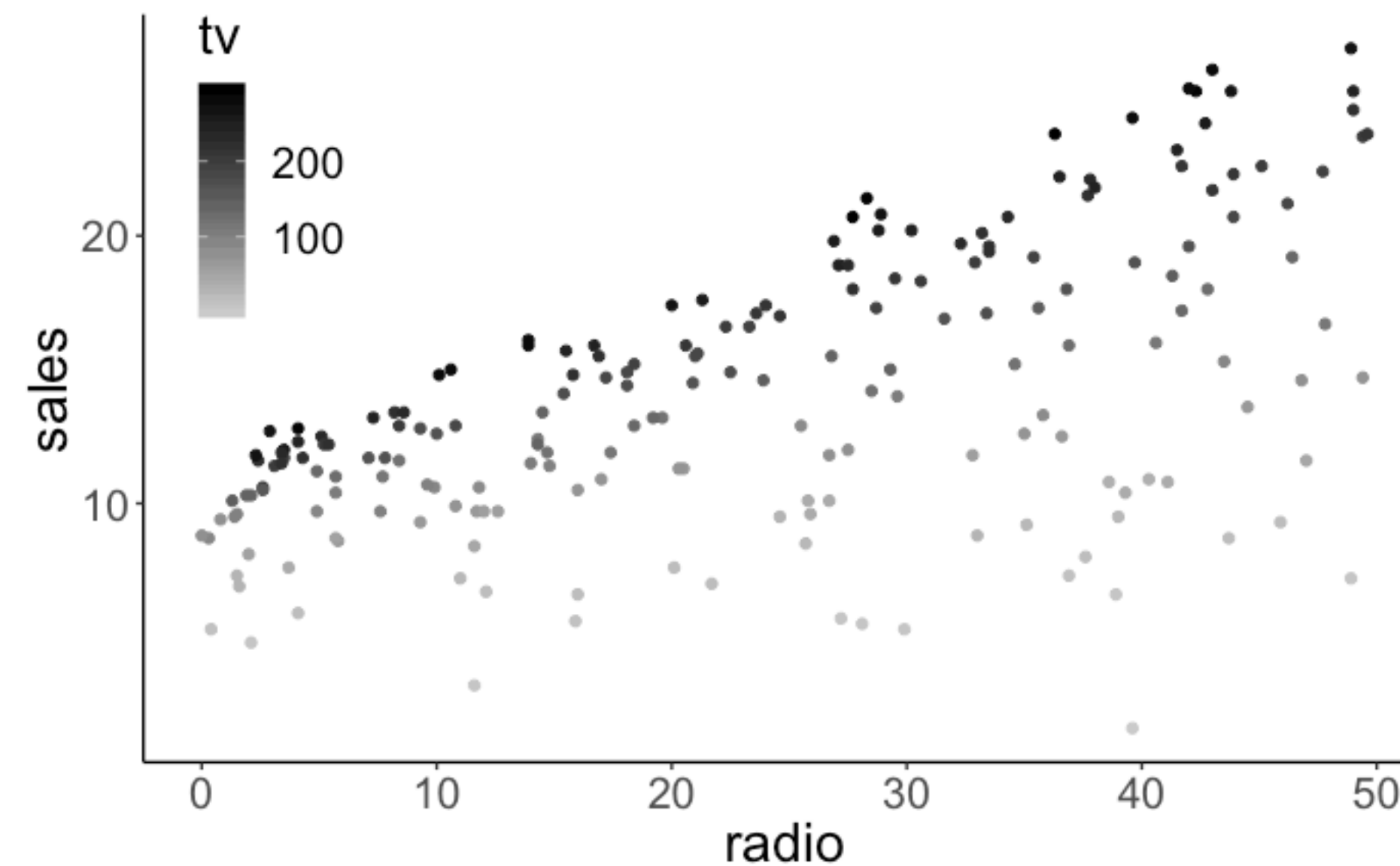
term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	2.92	0.29	9.92	0	2.34	3.50
tv	0.05	0.00	32.91	0	0.04	0.05
radio	0.19	0.01	23.38	0	0.17	0.20

$$\text{sales}_i = b_0 + b_1 \cdot \text{tv}_i + b_2 \cdot \text{radio}_i + e_i$$

$$\widehat{\text{sales}}_i = 2.92 + 0.05 \cdot \text{tv}_i + 0.19 \cdot \text{radio}_i$$

For a given amount of TV advertising, each additional \$1000 on radio advertising is associated with 190 unit higher sales.

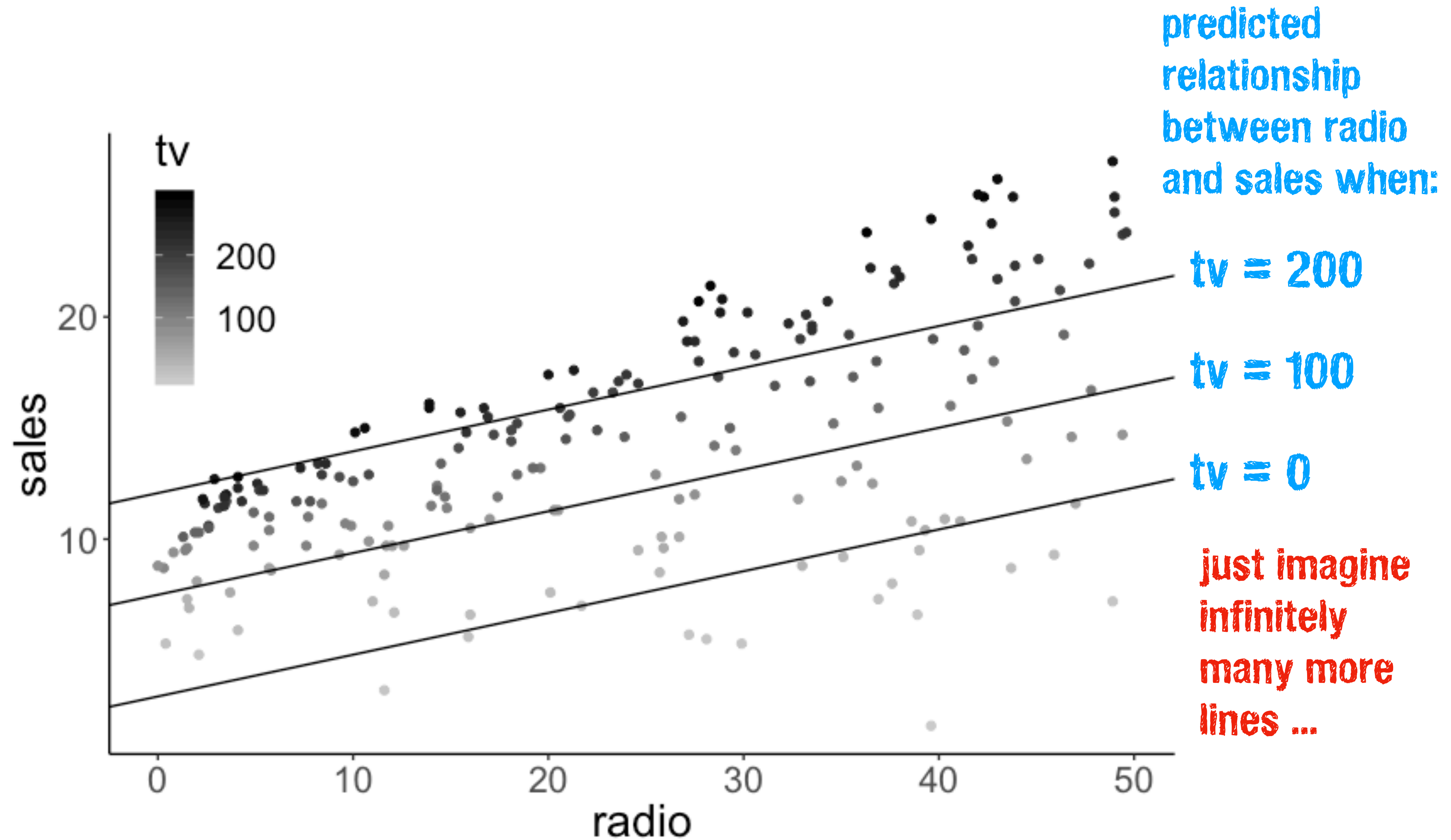
Reporting results



There is a significant relationship between sales and radio ads, controlling for TV ads $F(1, 197) = 546.74, p < .001$.

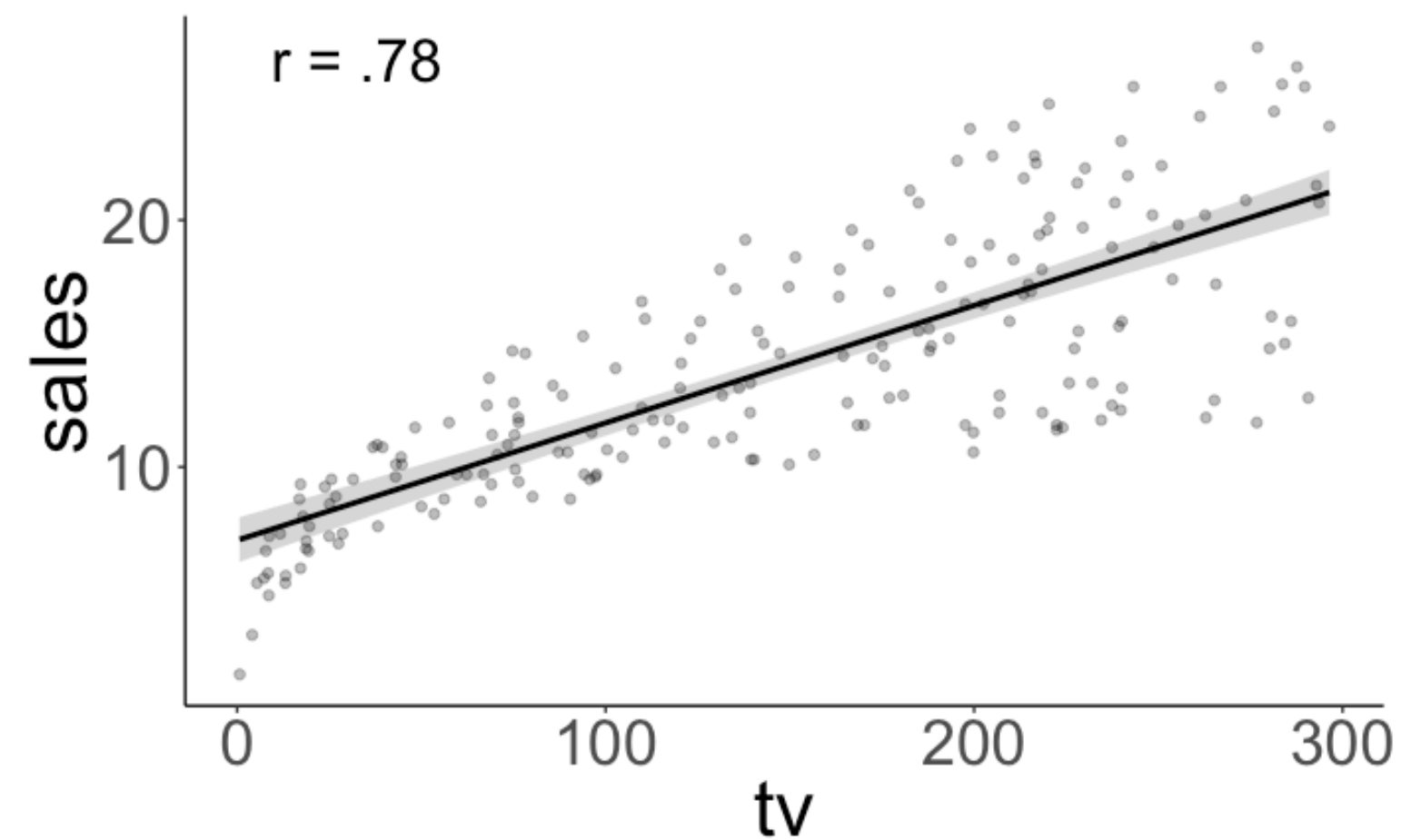
Holding TV ads fixed, each additional \$1000 spent on radio ads is associated with an additional sales of 190 units [170, 200] (95% confidence intervals).

Visualizing the results

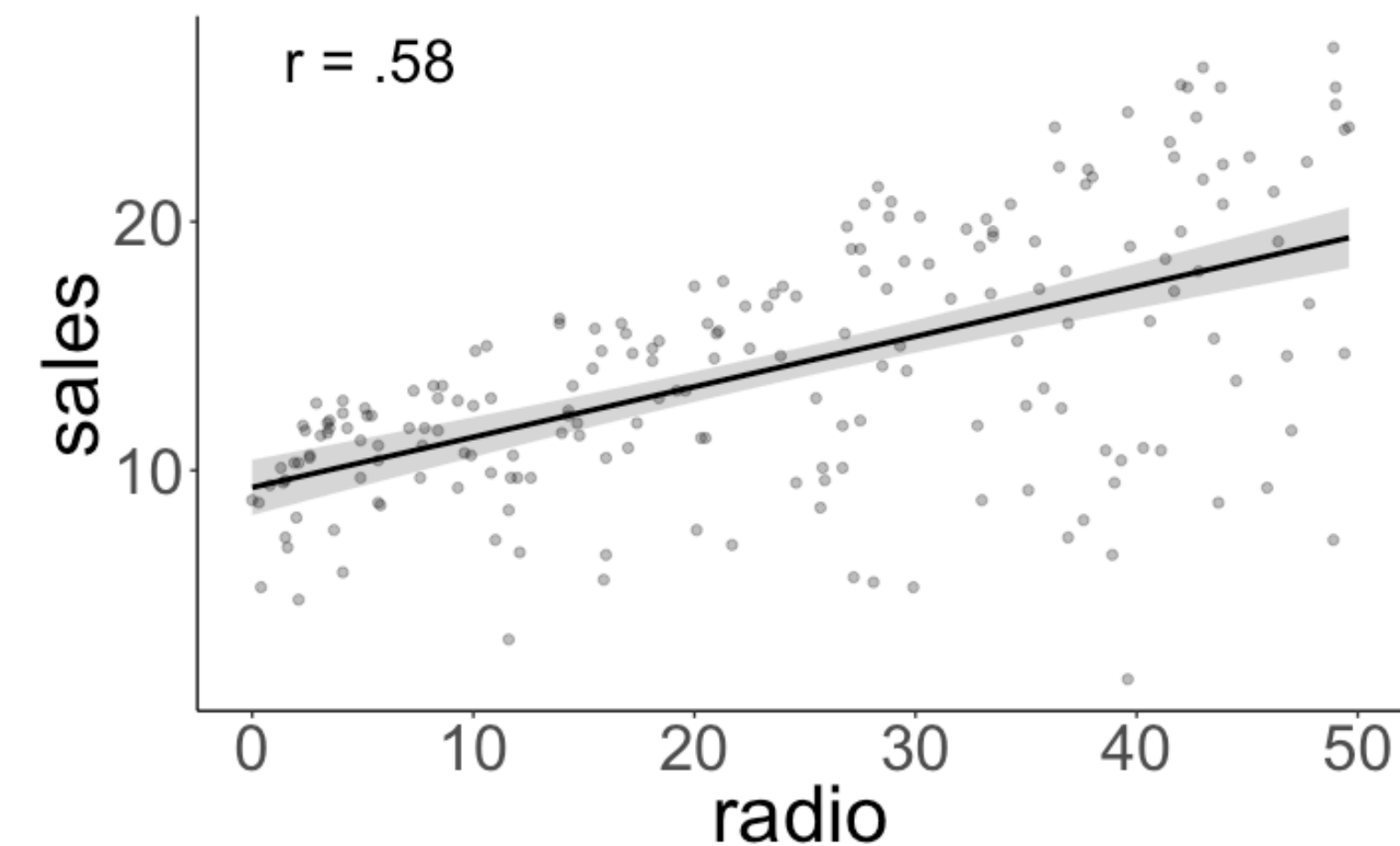


**Why can't we just run
several simple regressions?**

Relationship between TV ads and sales



Relationship between radio ads and sales



We found that both TV ads and radio ads were related to sales.

But did we need to run a multiple regression? Could we have just looked at correlations?

Advertising data set

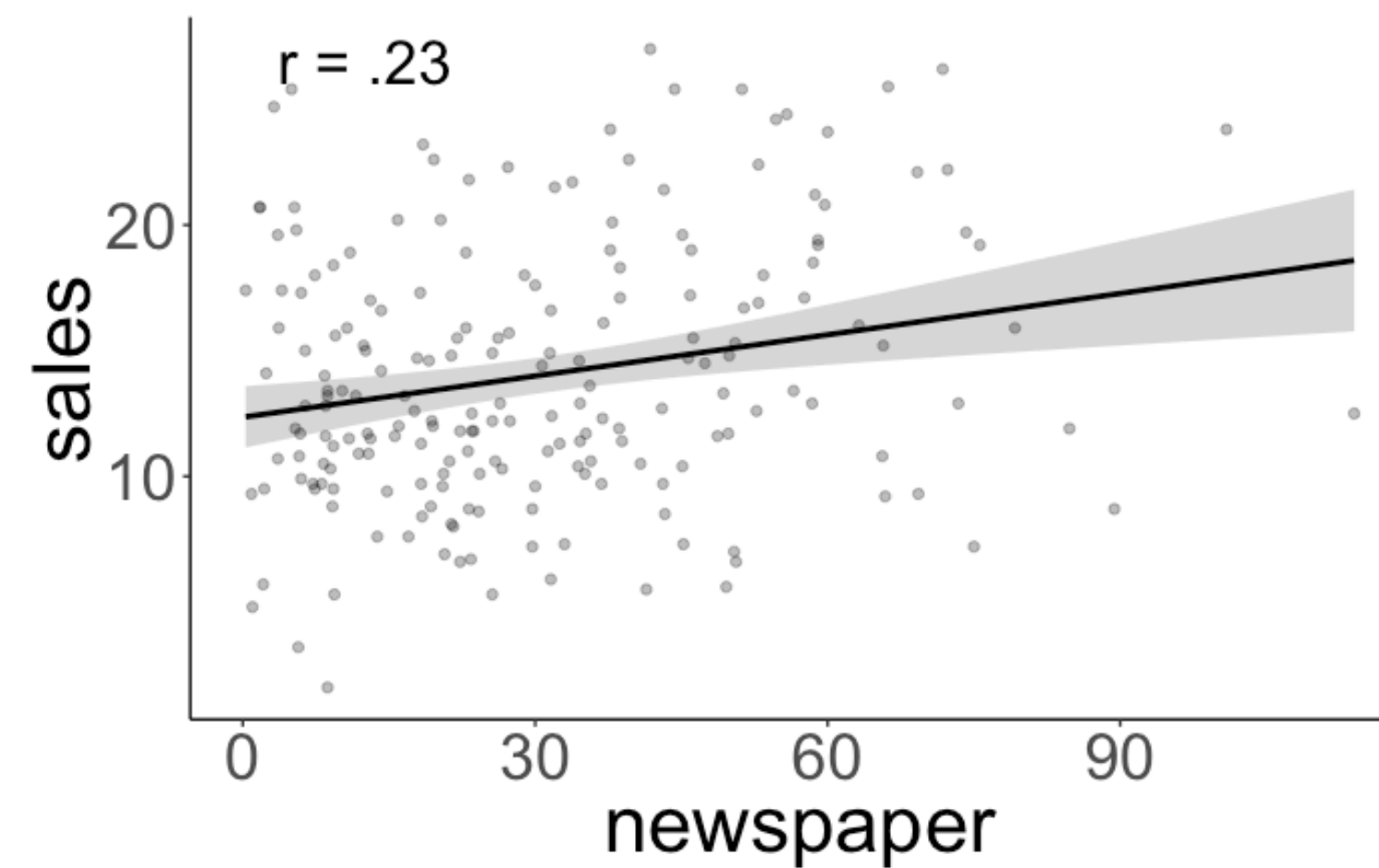
money spent on
different media
(x \$1000)

sales
(x1000)

index	tv	radio	newspaper	sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
3	17.2	45.9	69.3	9.3
4	151.5	41.3	58.5	18.5
5	180.8	10.8	58.4	12.9
6	8.7	48.9	75.0	7.2
7	57.5	32.8	23.5	11.8
8	120.2	19.6	11.6	13.2
9	8.6	2.1	1.0	4.8
10	199.8	2.6	21.2	10.6

Are newspaper ads and sales related when controlling for radio ads and TV ads?

Relationship between newspaper ads and sales



**this is
significant**


```

1 # fit the models
2 fit_c = lm(sales ~ 1 + tv + radio, data = df.ads)
3 fit_a = lm(sales ~ 1 + tv + radio + newspaper, data = df.ads)
4
5 # do the F test
6 anova(fit_c, fit_a)

```

Analysis of Variance Table

Model 1: sales ~ 1 + tv + radio

Model 2: sales ~ 1 + tv + radio + newspaper

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	197	556.91				
2	196	556.83	1	0.088717	0.0312	0.8599

it's not worth it

sales ~ 1 + tv

sales ~ 1 + tv + newspaper

it's worth it

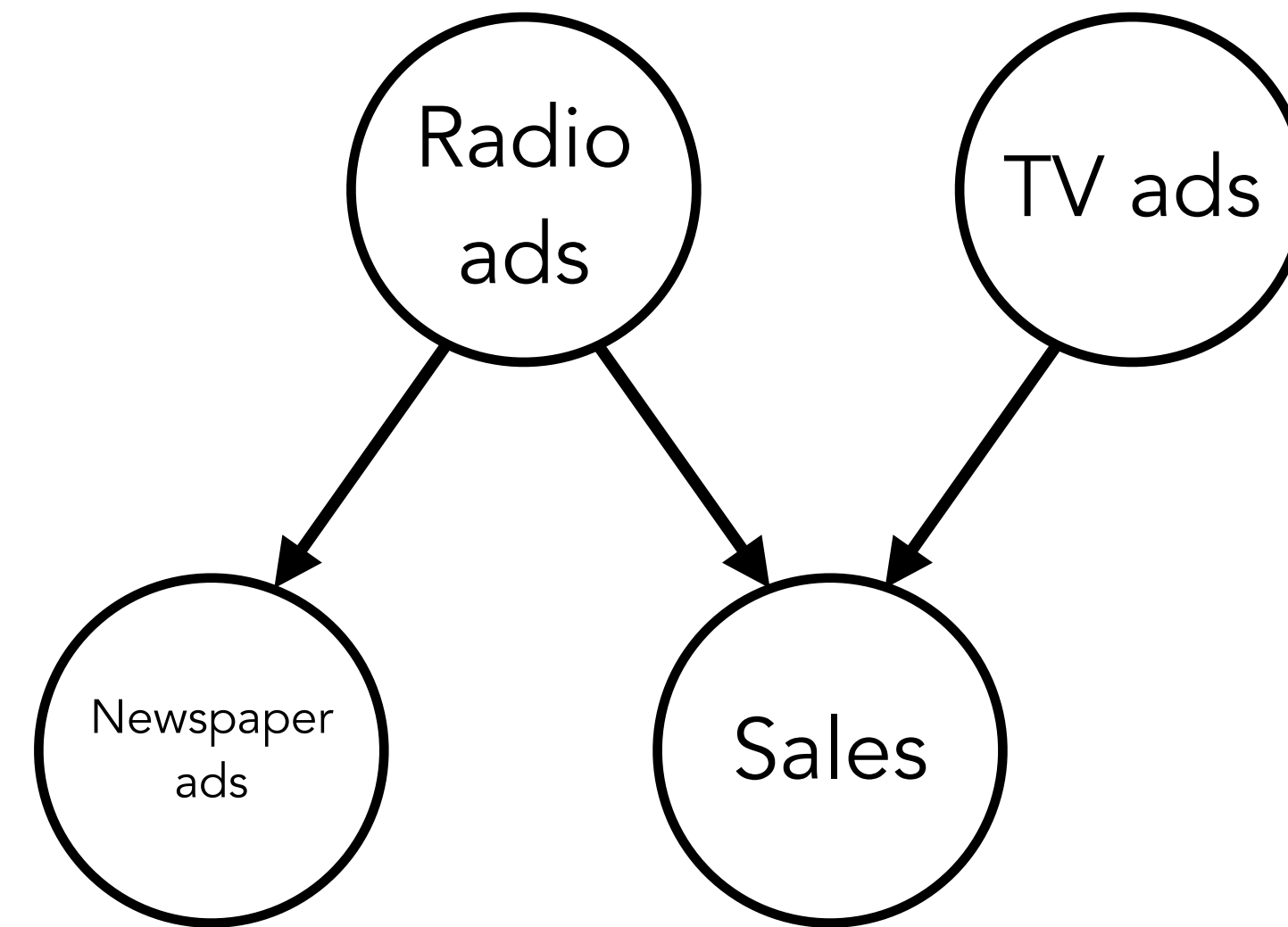
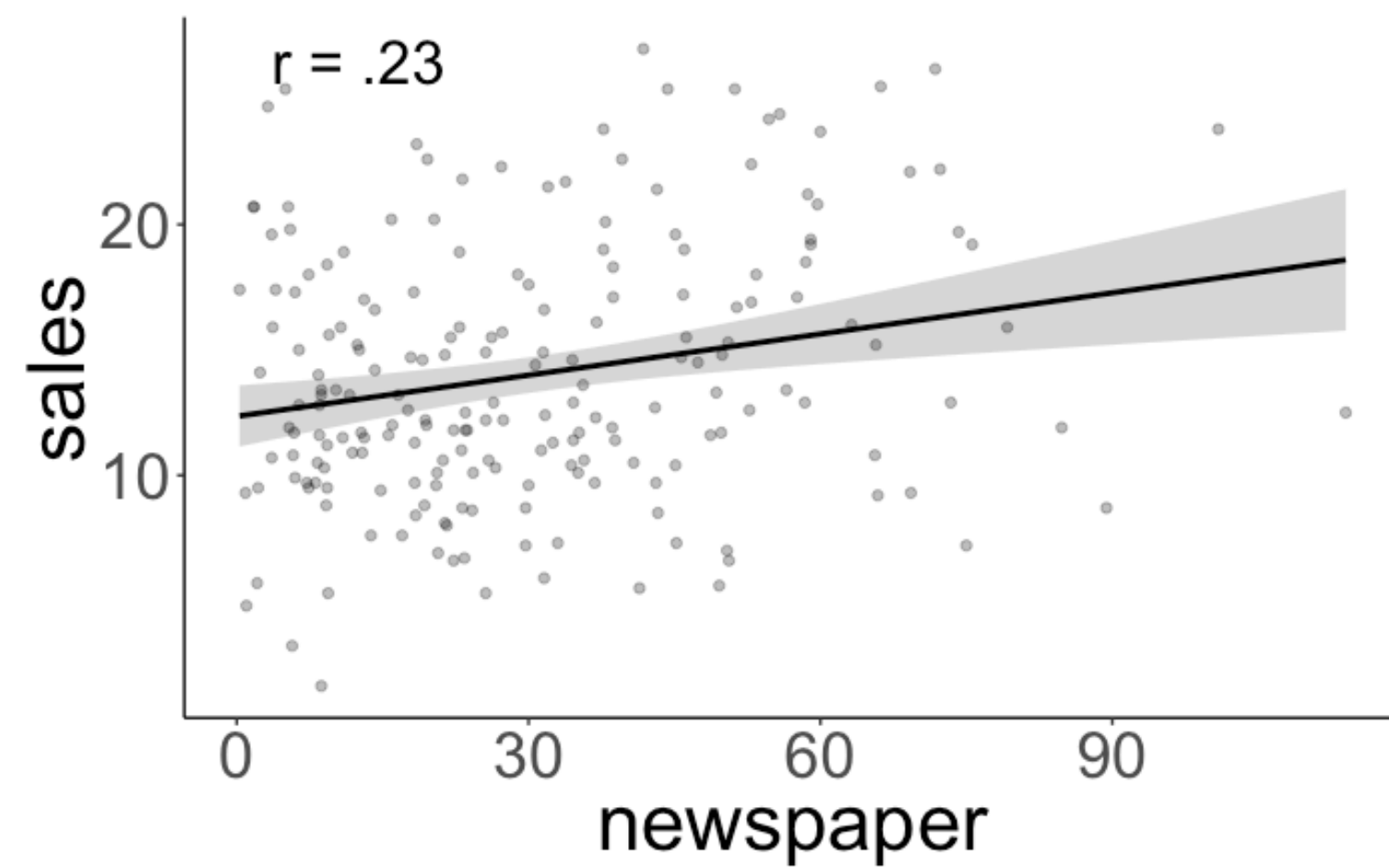
sales ~ 1 + radio

sales ~ 1 + radio + newspaper

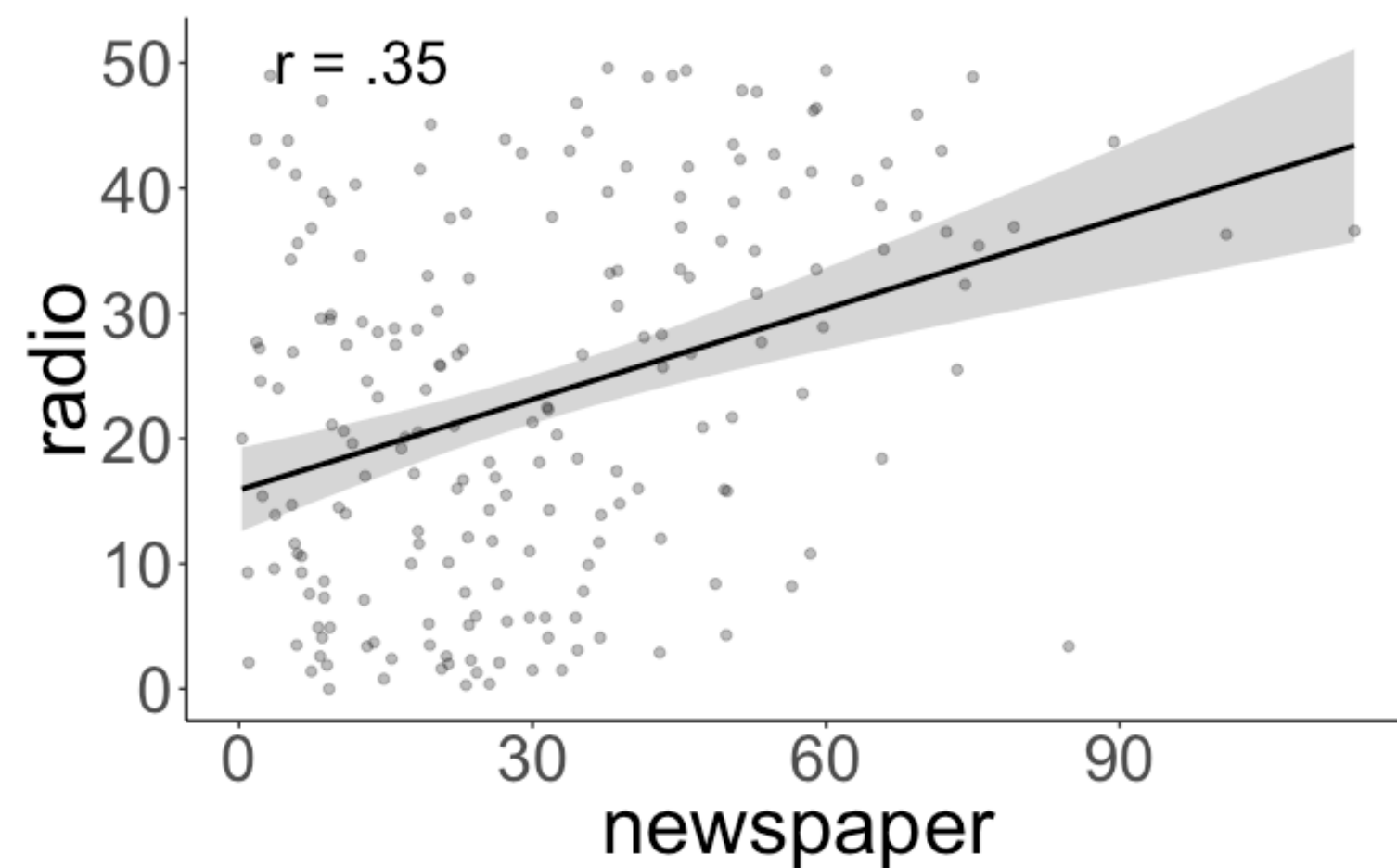
it's not worth it

Are newspaper ads and sales related when controlling for radio ads and TV ads?

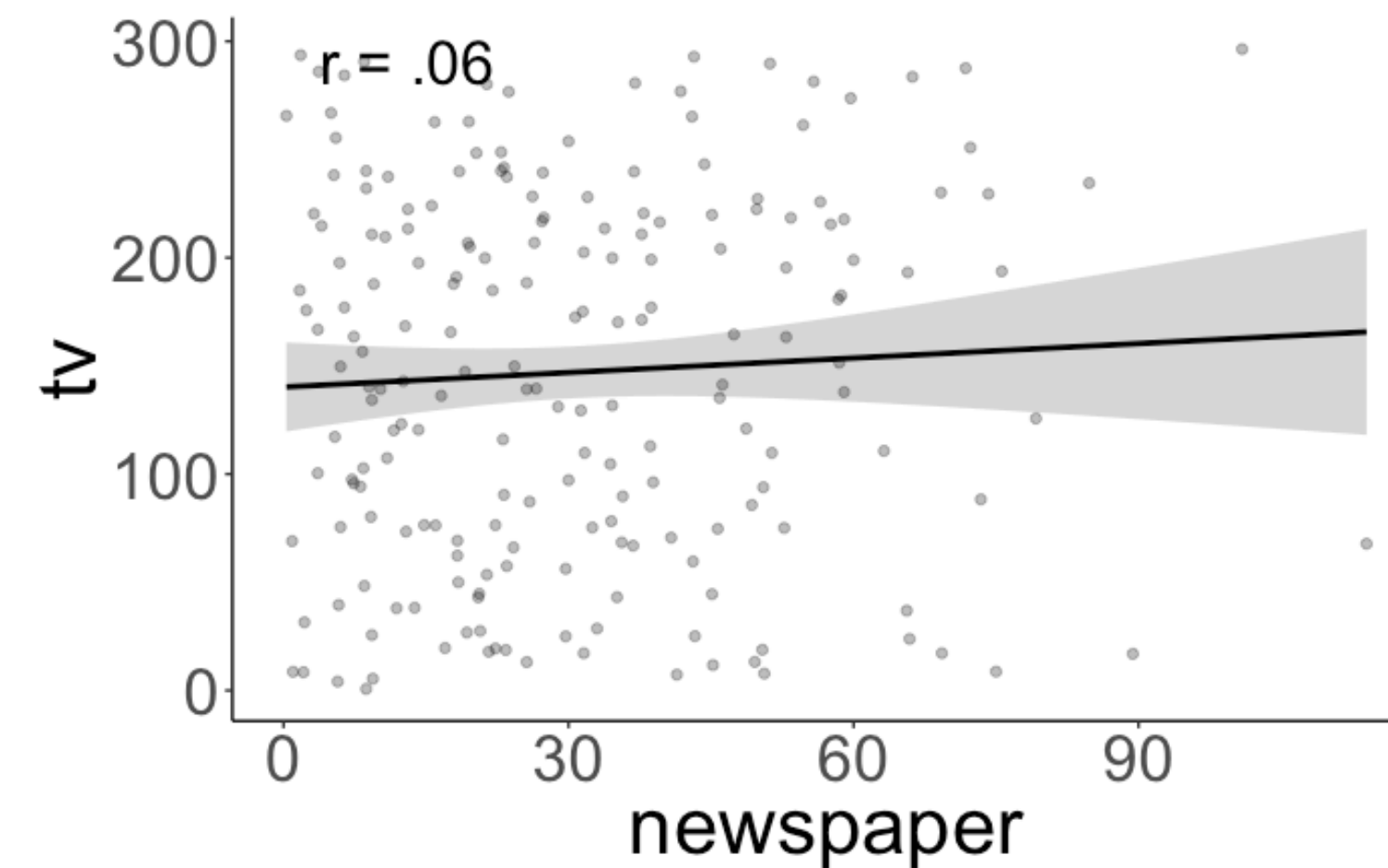
Relationship between newspaper ads and sales



Relationship between newspaper and radio ads



Relationship between newspaper and TV ads



Categorical predictors

Credit data set

Categorical/Factor
Variable

Continuous
Variable

df.credit

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
1	14.89	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.03	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.59	7075	514	4	71	11	Male	No	No	Asian	580
4	148.92	9504	681	3	36	11	Female	No	No	Asian	964
5	55.88	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	21.00	3388	259	2	37	12	Female	No	No	African American	203
8	71.41	7114	512	2	87	9	Male	No	No	Asian	872
9	15.12	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.06	6819	491	3	41	19	Female	Yes	Yes	African American	1350

variable	description
income	in thousand dollars
limit	credit limit
rating	credit rating
cards	number of credit cards
age	in years
education	years of education
gender	male or female
student	student or not
married	married or not
ethnicity	African American, Asian, Caucasian
balance	average credit card debt in dollars

nrow(df.credit) = 400

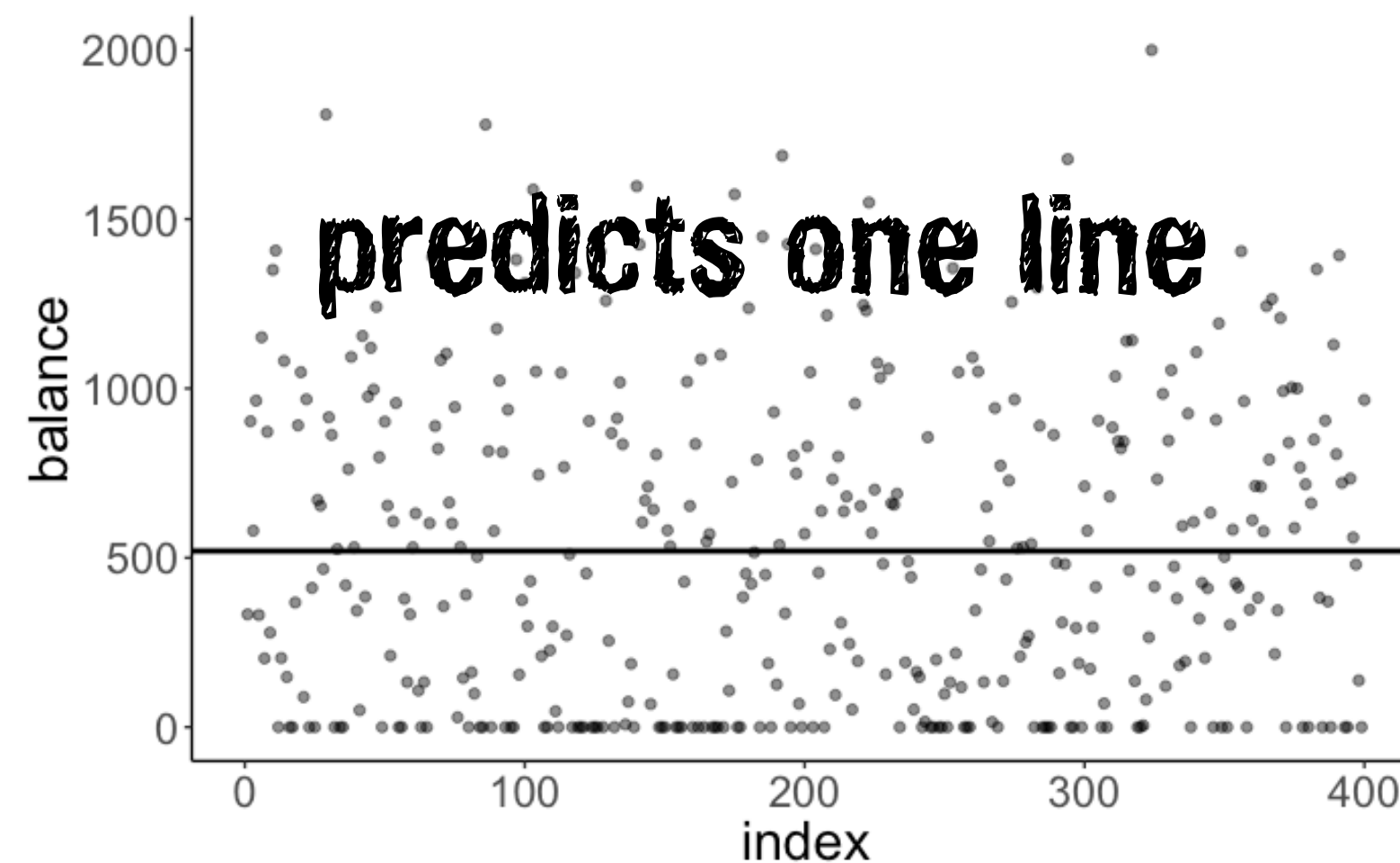
Do students have a different average credit card balance from non-students?

H_0 : Students and non-students have the same balance.

Model C

$$Y_i = \beta_0 + \epsilon_i$$

Model prediction



Fitted model

$$Y_i = 520.02 + e_i$$

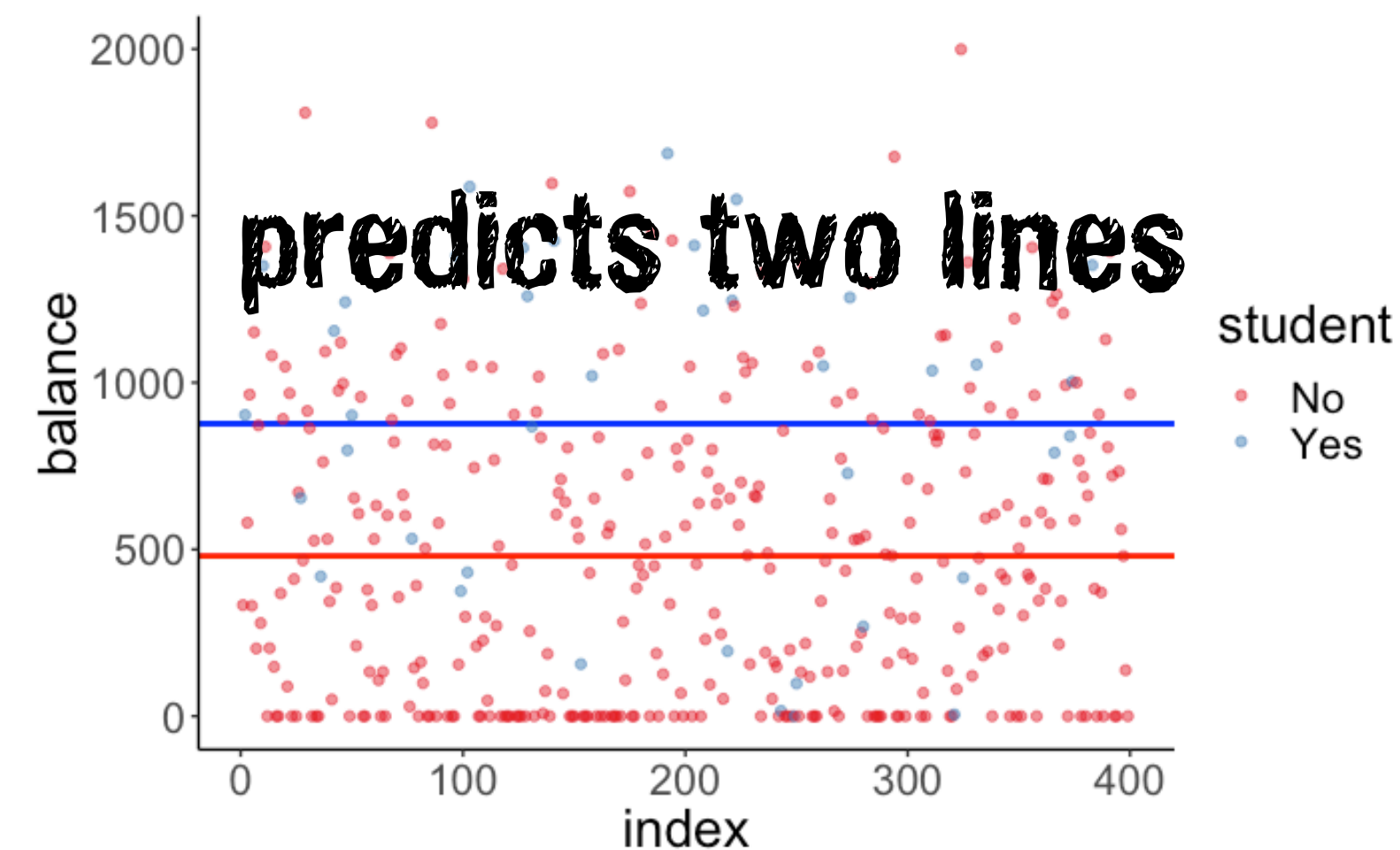
H_1 : Students and non-students have different balances.

Model A

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

student (2 categories)

Model prediction



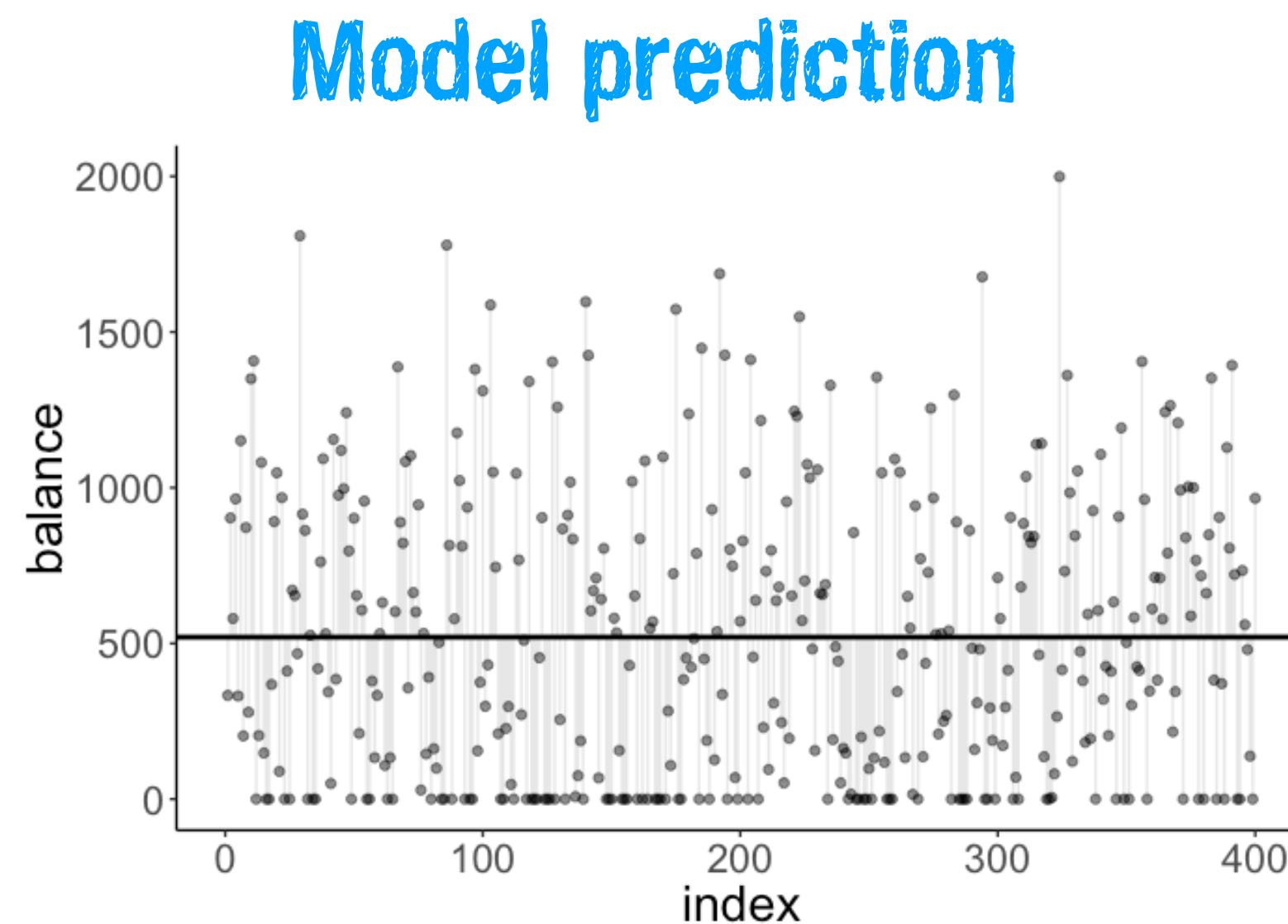
Fitted model

$$Y_i = 480.37 + 396.46X_i + e_i$$

H_0 : Students and non-students have the same balance.

Model C

$$Y_i = \beta_0 + \epsilon_i$$



Fitted model

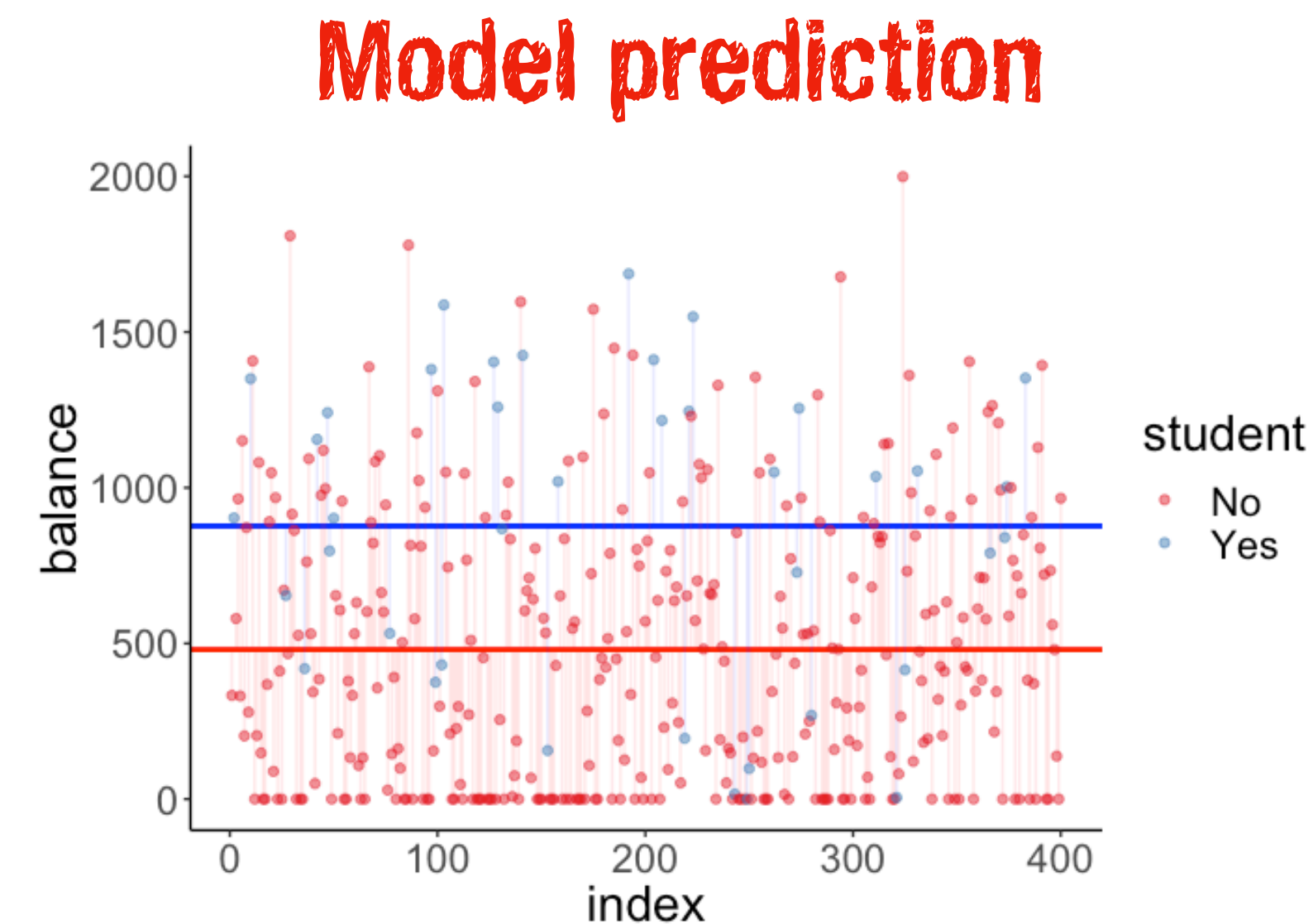
$$Y_i = 520.02 + e_i$$

H_1 : Students and non-students have different balances.

Model A

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

student (2 categories)



Fitted model

$$Y_i = 480.37 + 396.46X_i + e_i$$

Worth it?

```
1 # fit the models
2 fit_c = lm(balance ~ 1, data = df.credit)
3 fit_a = lm(balance ~ 1 + student, data = df.credit)
4
5 # run the F test
6 anova(fit_c, fit_a)
```

Analysis of Variance Table

Model 1: balance ~ 1

Model 2: balance ~ student

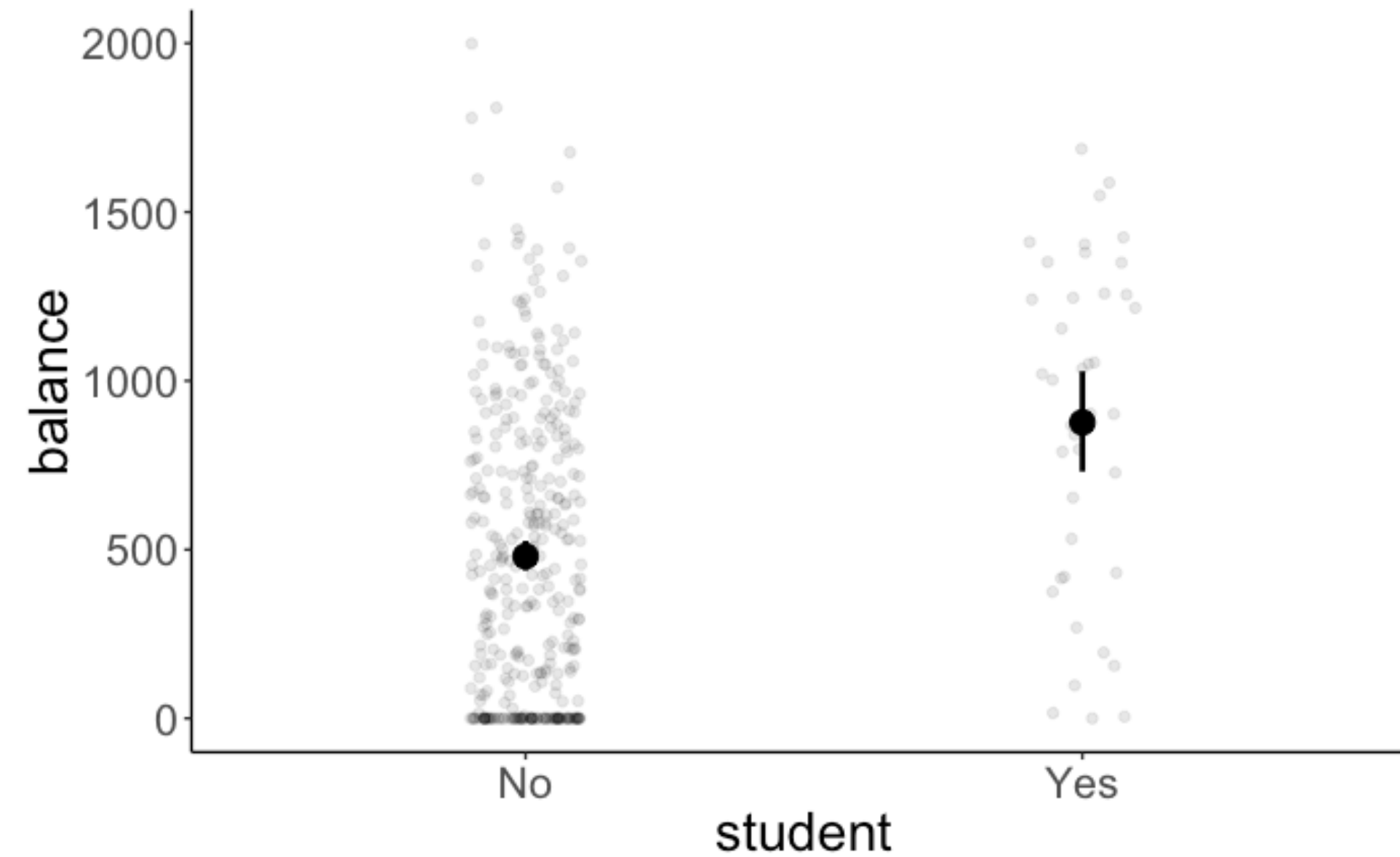
	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	399	84339912				
2	398	78681540	1	5658372	28.622	1.488e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Worth it!

Two sample t-test (with independent groups)

Reporting the results



Students have a significantly higher average credit card balance ($Mean = 876.83$, $SD = 490.00$) than non-students ($Mean = 480.37$, $SD = 439.41$), $F(1, 398) = 28.622$, $p < .001$.

Interpreting the model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

```
1 fit_a = lm(balance ~ 1 + student, data = df.credit)
2 fit_a %>%
3   summary()
```

Call:

```
lm(formula = balance ~ student, data = df.credit)
```

Residuals:

Min	1Q	Median	3Q	Max
-876.82	-458.82	-40.87	341.88	1518.63

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	480.37	23.43	20.50	< 2e-16 ***
studentYes	396.46	74.10	5.35	1.49e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 444.6 on 398 degrees of freedom

Multiple R-squared: 0.06709, Adjusted R-squared: 0.06475

F-statistic: 28.62 on 1 and 398 DF, p-value: 1.488e-07

Dummy coding



Dummy coding

$$\hat{Y}_i = 480.37 + 396.46 \cdot \text{student_dummy}_i$$

if student = "No" $\hat{Y}_i = 480.37$

if student = "Yes" $\hat{Y}_i = 480.37 + 396.46 = 876.83$

student	student_dummy
No	0
Yes	1
No	0
No	0
No	0
No	0
No	0
No	0
No	0
Yes	1

- Reference category is coded as 0, the other category is coded as 1
- When thrown into an `lm()`, R automatically turns character columns into factors, and determines the reference category alphabetically

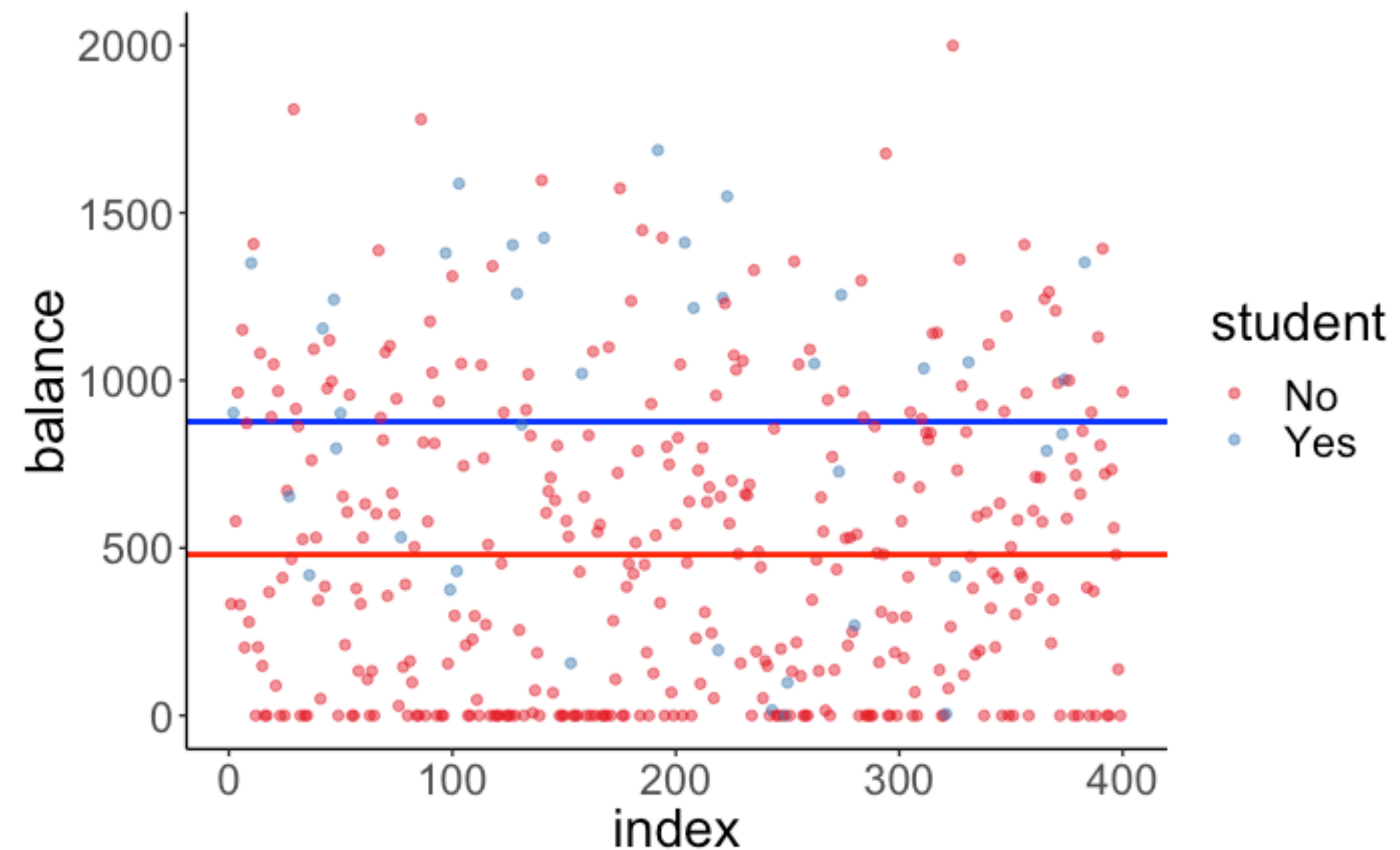
Dummy coding

$$\hat{Y}_i = 480.37 + 396.46 \cdot \text{student_dummy}_i + 0_i$$

if student = "No" $\hat{Y}_i = 480.37$

if student = "Yes" $\hat{Y}_i = 480.37 + 396.46 = 876.83$

student	student_dummy
No	0
Yes	1
No	0
No	0
No	0
No	0
No	0
No	0
Yes	1



Categorical and continuous predictor

Credit data set

Continuous Variable

Categorical/Factor Variable

Continuous Variable

df.credit

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
1	14.89	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.03	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.59	7075	514	4	71	11	Male	No	No	Asian	580
4	148.92	9504	681	3	36	11	Female	No	No	Asian	964
5	55.88	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	21.00	3388	259	2	37	12	Female	No	No	African American	203
8	71.41	7114	512	2	87	9	Male	No	No	Asian	872
9	15.12	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.06	6819	491	3	41	19	Female	Yes	Yes	African American	1350

variable	description
income	in thousand dollars
limit	credit limit
rating	credit rating
cards	number of credit cards
age	in years
education	years of education
gender	male or female
student	student or not
married	married or not
ethnicity	African American, Asian, Caucasian
balance	average credit card debt in dollars

nrow(df.credit) = 400

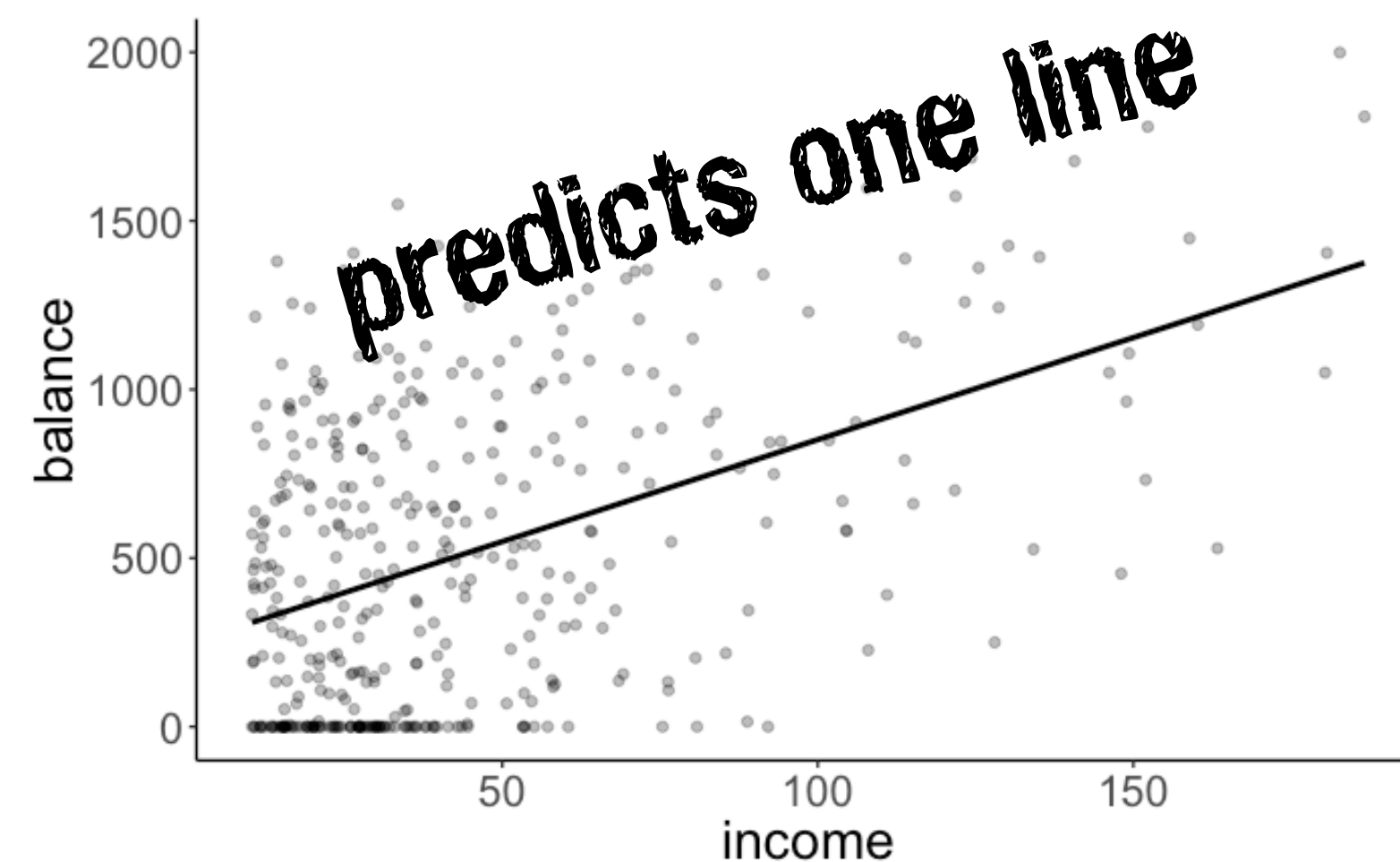
Do students have a different average credit card balance from non-students, when controlling for income?

H_0 : Students and non-students have the same balance, when controlling for income.

Model C

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i + \epsilon_i$$

Model prediction



Fitted model

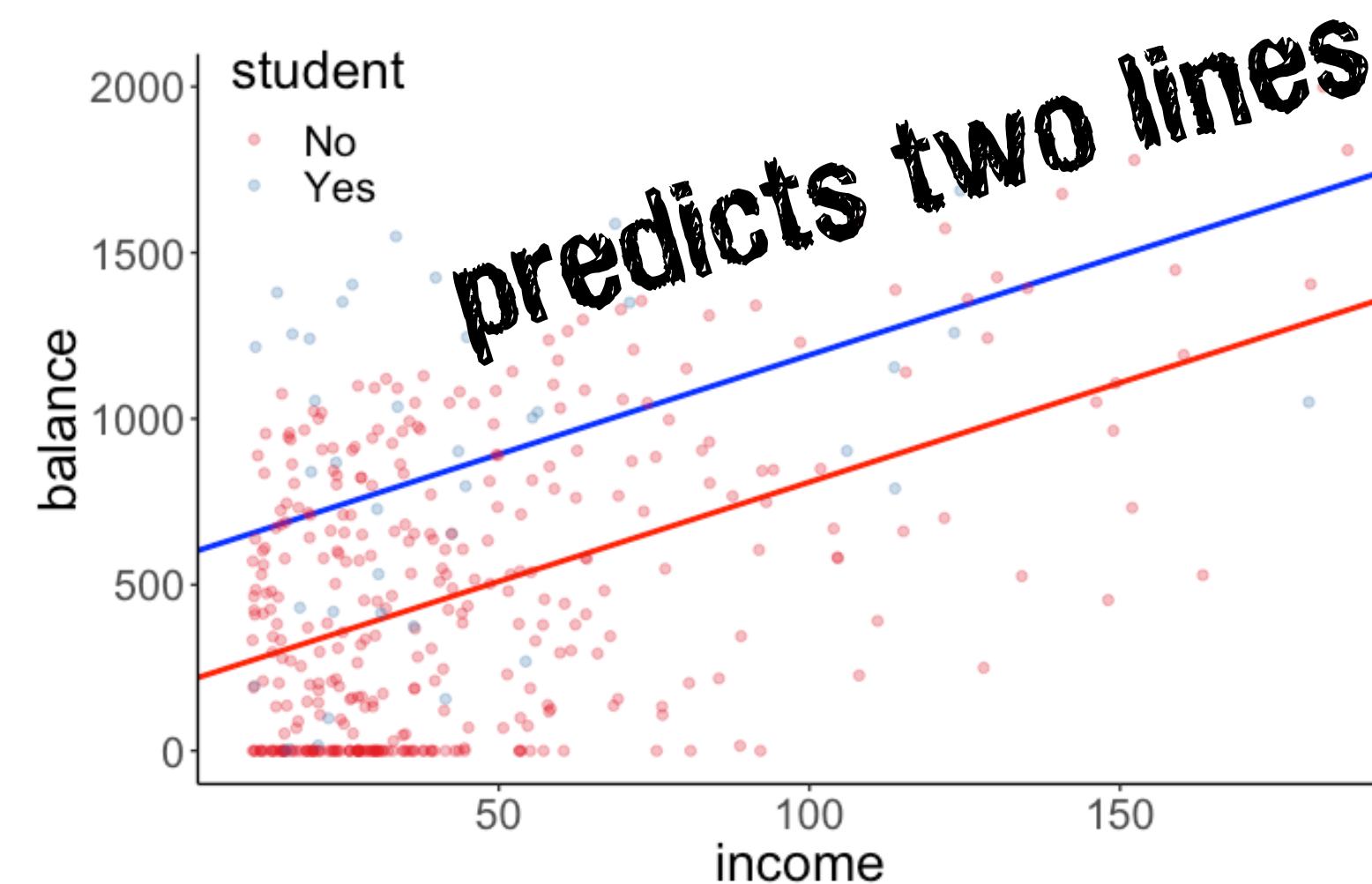
$$\widehat{\text{balance}}_i = 246.515 + 6.048 \cdot \text{income}_i + 0$$

H_1 : Students and non-students have different balances, when controlling for income.

Model A

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{student}_i + \epsilon_i$$

Model prediction



Fitted model

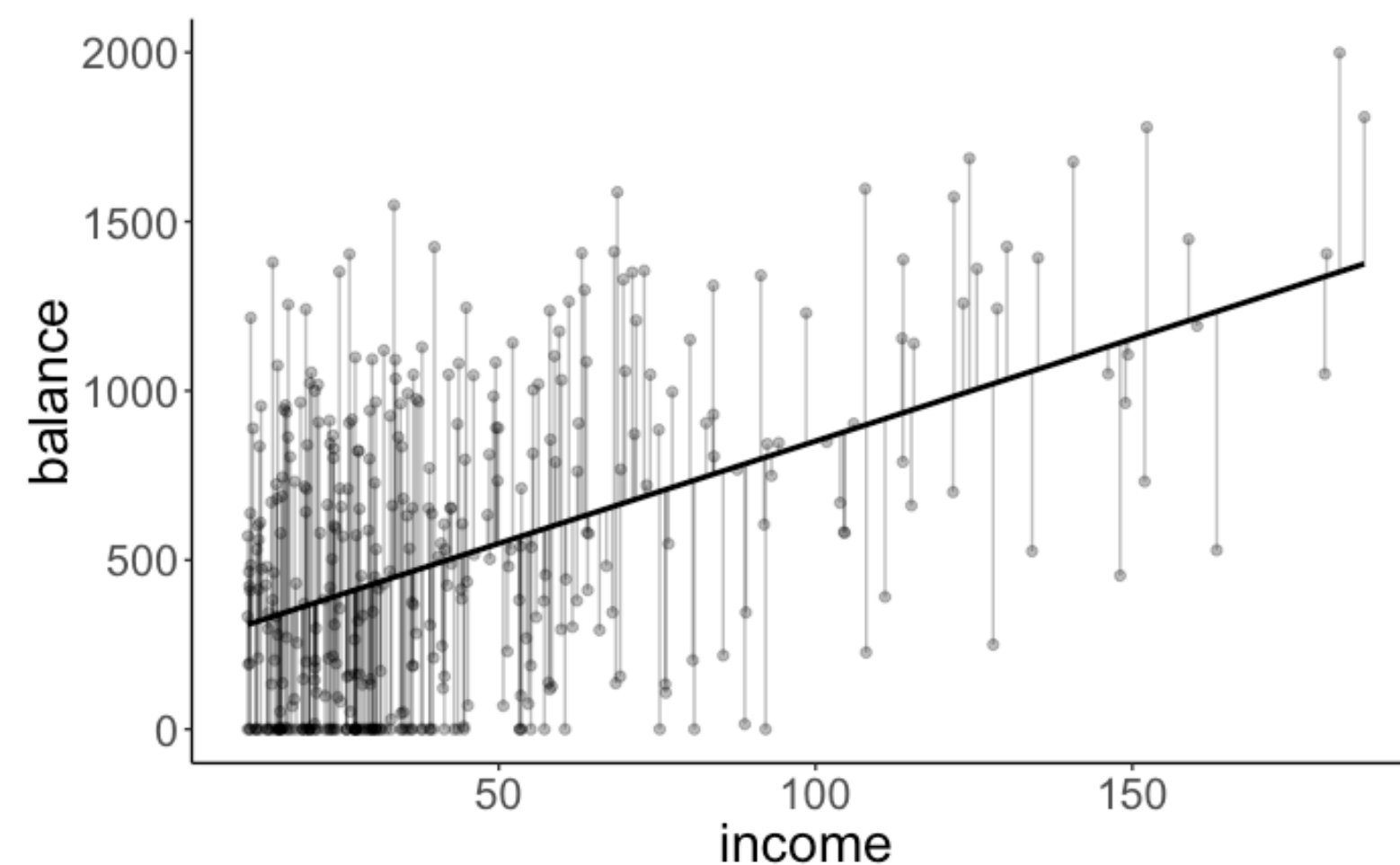
$$\widehat{\text{balance}}_i = 211.14 + 5.98 \cdot \text{income}_i + 382.67 \cdot \text{student}_i + 0$$

H_0 : Students and non-students have the same balance, when **controlling for income**.

Model C

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i + \epsilon_i$$

Model prediction



Fitted model

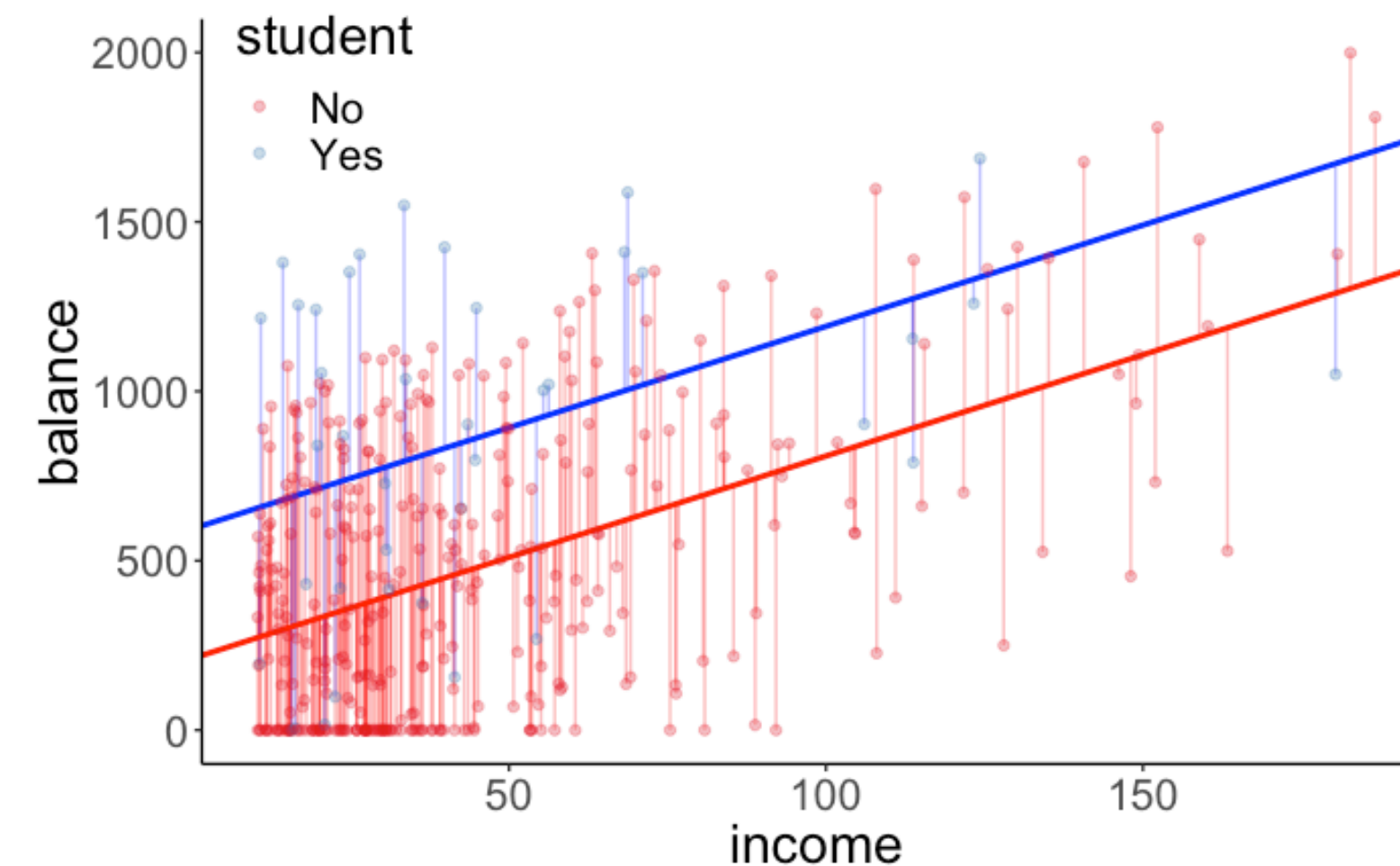
$$\widehat{\text{balance}}_i = 246.515 + 6.048 \cdot \text{income}_i + 0_i$$

H_1 : Students and non-students have different balances, when **controlling for income**.

Model A

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{student}_i + \epsilon_i$$

Model prediction



Fitted model

$$\widehat{\text{balance}}_i = 211.14 + 5.98 \cdot \text{income}_i + 382.67 \cdot \text{student}_i + 0_i$$

Worth it?

```
1 # fit the models
2 fit_c = lm(balance ~ 1 + income, df.credit)
3 fit_a = lm(balance ~ 1 + income + student, df.credit)
4
5 # run the F test
6 anova(fit_c, fit_a)
```

Analysis of Variance Table

Model 1: balance ~ 1 + income

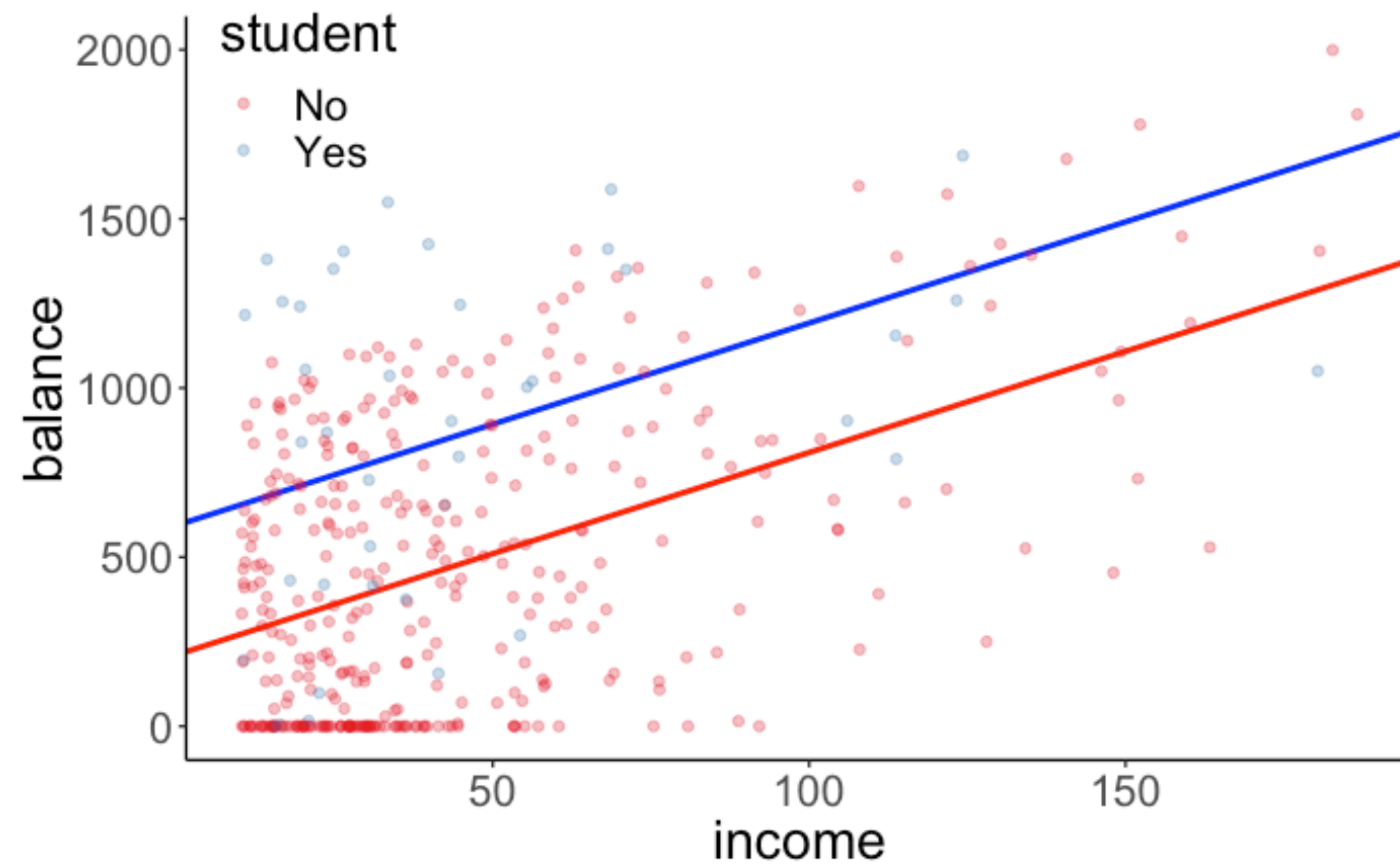
Model 2: balance ~ 1 + income + student

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	398	66208745				
2	397	60939054	1	5269691	34.331	9.776e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Worth it!

Interpretation

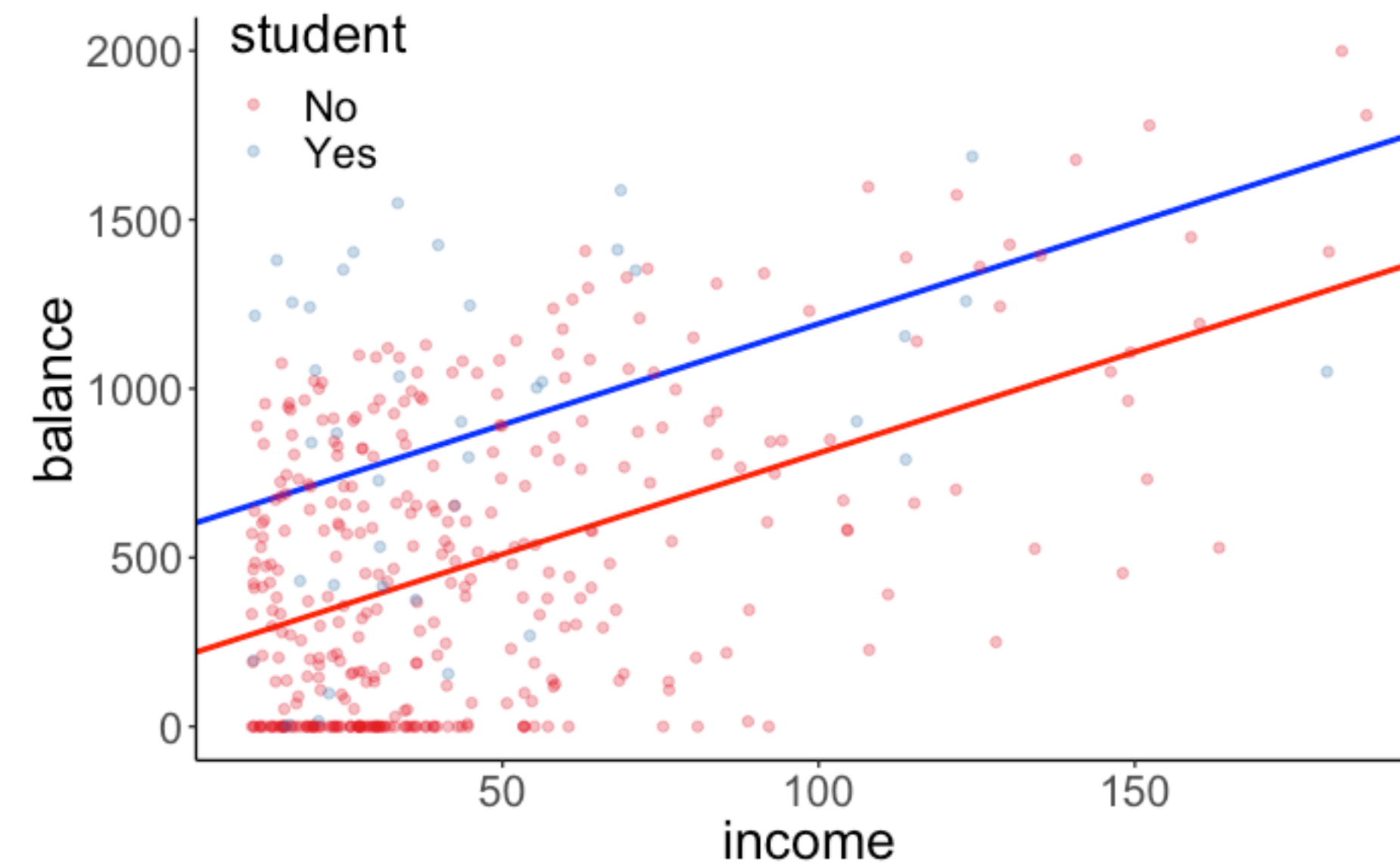


$$\widehat{\text{balance}}_i = 211.14 + 5.98 \cdot \text{income}_i + 382.67 \cdot \text{student}_i + 0$$

if **student = "No"** $\widehat{\text{balance}}_i = 211.14 + 5.98 \cdot \text{income}_i + (382.67 \cdot 0)$

if **student = "Yes"** $\widehat{\text{balance}}_i = 211.14 + 5.98 \cdot \text{income}_i + (382.67 \cdot 1)$
 $= 211.14 + 382.67 + 5.98 \cdot \text{income}_i$
 $= 593.81 + 5.98 \cdot \text{income}_i$

Reporting the results



Controlling for income, students have a significantly higher average credit card balance ($Mean = 876.83, SD = 490.00$) than non-students ($Mean = 480.37, SD = 439.41$), $F(1, 397) = 34.331, p < .001$.

Interactions

Is the ***relationship*** between level of income and balance **different** for students than it is for non-students?

Compact Model

$$\widehat{\text{balance}}_i = b_0 + b_1 \text{income}_i + b_2 \text{student}_i + \epsilon_i$$

**R-formula
for these?**

Augmented Model

$$\widehat{\text{balance}}_i = b_0 + b_1 \text{income}_i + b_2 \text{student}_i + b_3 (\text{income}_i \times \text{student}_i) + \epsilon_i$$

Interaction ~ Moderation

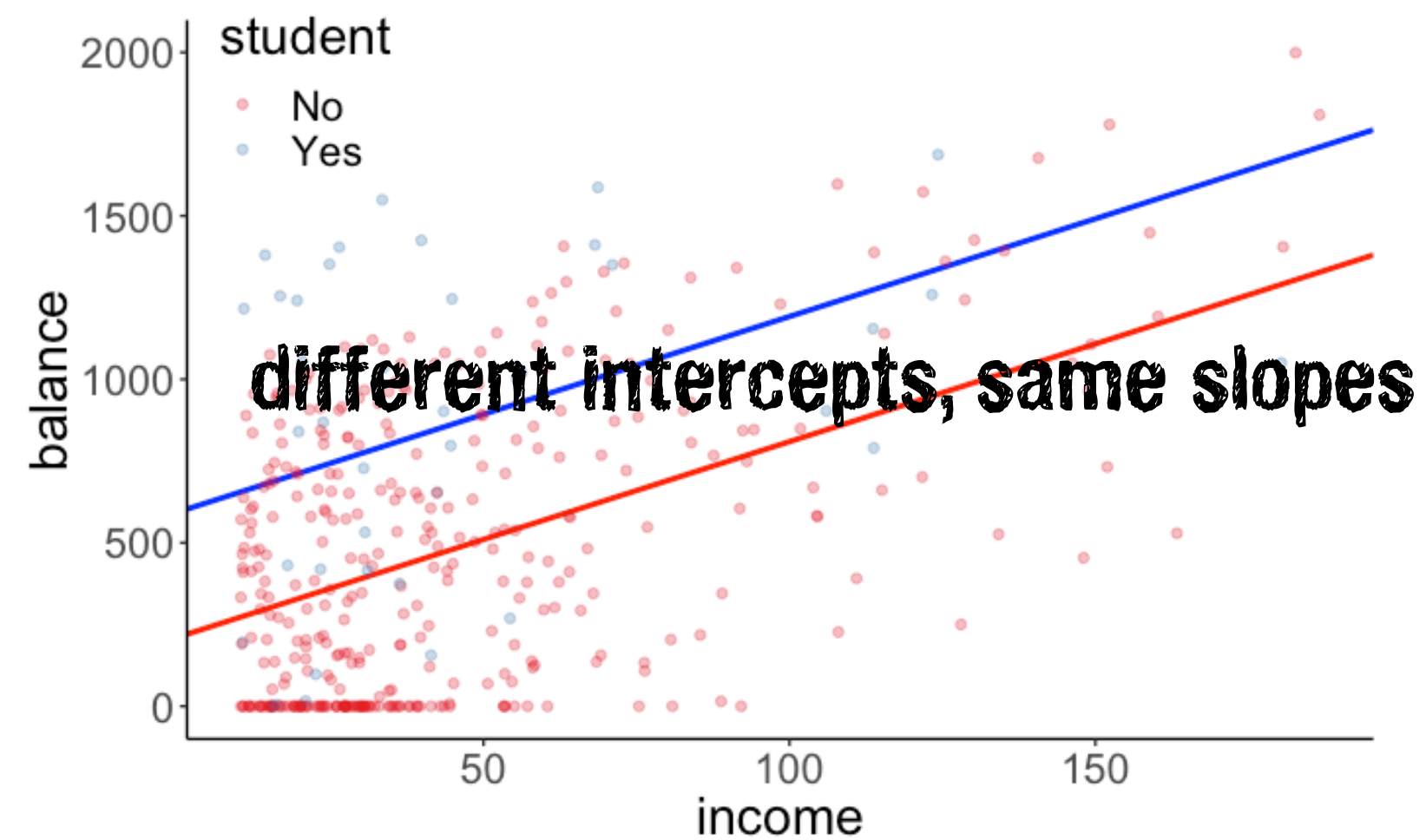
Does student status moderate the relationship between level of income and balance?

H_0 : The relationship between income and balance is the same for students and non-students.

Model C

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i + \beta_2 \text{student}_i + \epsilon_i$$

Model prediction



Fitted model

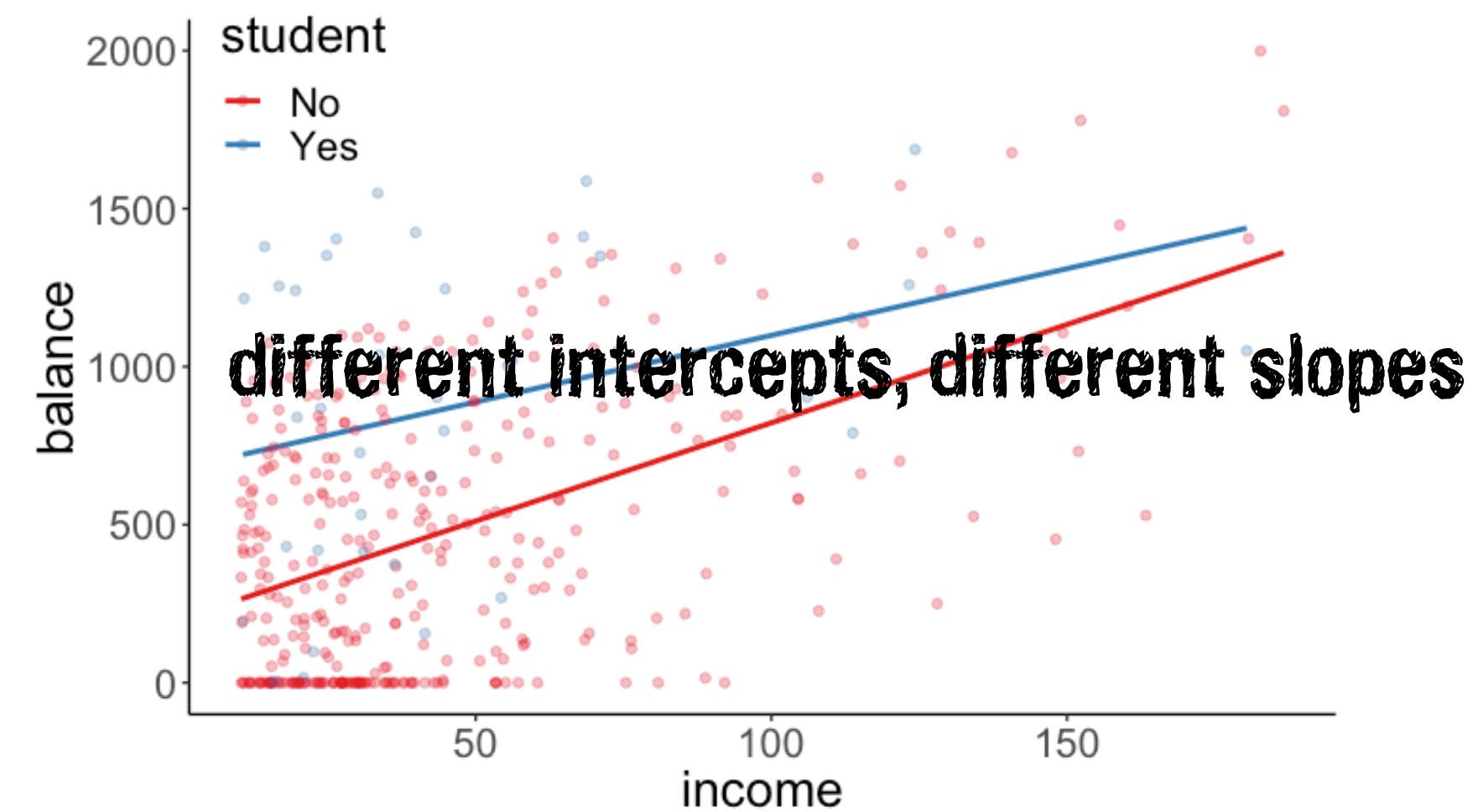
$$\widehat{\text{balance}}_i = 211.14 + 5.98 \cdot \text{income}_i + 382.67 \cdot \text{student}_i$$

H_1 : The relationship between income and balance differs between students and non-students.

Model A

$$\widehat{\text{balance}}_i = b_0 + b_1 \text{income}_i + b_2 \text{student}_i + b_3 (\text{income}_i \times \text{student}_i) + \epsilon_i$$

Model prediction



Fitted model

$$\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot \text{student}_i - 2.00 \cdot (\text{income}_i \times \text{student}_i)$$

Worth it?

Is the relationship between level of income and balance different for students than it is for non-students?

```
1 # fit models
2 fit_c = lm(formula = balance ~ 1 + income + student, data = df.credit)
3 fit_a = lm(formula = balance ~ 1 + income * student, data = df.credit)
4
5 # F-test
6 anova(fit_c, fit_a)
```

Equations
for these?

Analysis of Variance Table **not worth it!**

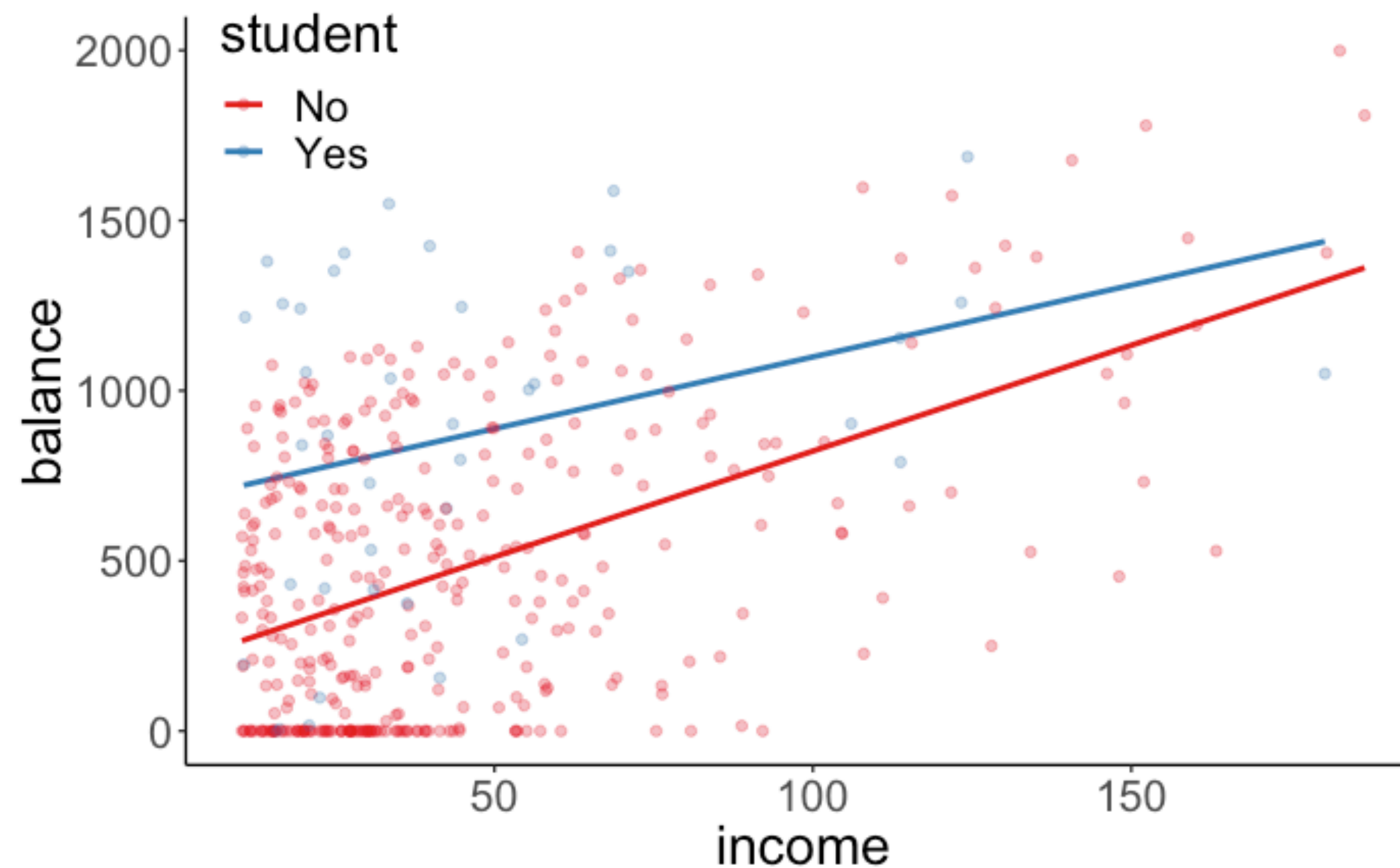
Model 1: balance ~ income + student

Model 2: balance ~ income * student

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	397	60939054				
2	396	60734545	1	204509	1.3334	0.2489

```
1 # alternative coding for model a
2 fit_a = lm(formula = balance ~ 1 + income + student + income:student, data = df.credit)
```

Interpretation



$$\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot \text{student}_i - 2.00 \cdot (\text{income}_i \times \text{student}_i) + 0$$

if student = "No" $\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i + (476.68 \cdot 0) - 2.00 \cdot (\text{income}_i \times 0)$

if student = "Yes"

$$\begin{aligned}\widehat{\text{balance}}_i &= 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot 1 - 2.00 \cdot (\text{income}_i \times 1) \\ &= 677.3 + 6.22 \cdot \text{income}_i - 2.00 \cdot \text{income}_i \\ &= 677.3 + 4.22 \cdot \text{income}_i\end{aligned}$$

Interpretation

```
fit1 = lm(formula = balance ~ income + student + income:student, data = df.credit)
```

Explicitly encode the interaction

```
1 df.credit %>%  
2   mutate(student_dummy = ifelse(student == "No", 0, 1)) %>%  
3   mutate(income_student = income * student_dummy) %>%  
4   select(balance, income, student, student_dummy, income_student)
```

balance	income	student	student_dummy	income_student
333	14.89	No	0	0.00
903	106.03	Yes	1	106.03
580	104.59	No	0	0.00
964	148.92	No	0	0.00
331	55.88	No	0	0.00
1151	80.18	No	0	0.00
203	21.00	No	0	0.00
872	71.41	No	0	0.00
279	15.12	No	0	0.00
1350	71.06	Yes	1	71.06

```
fit2 = lm(formula = balance ~ income + student + income_student, data = df.credit)
```

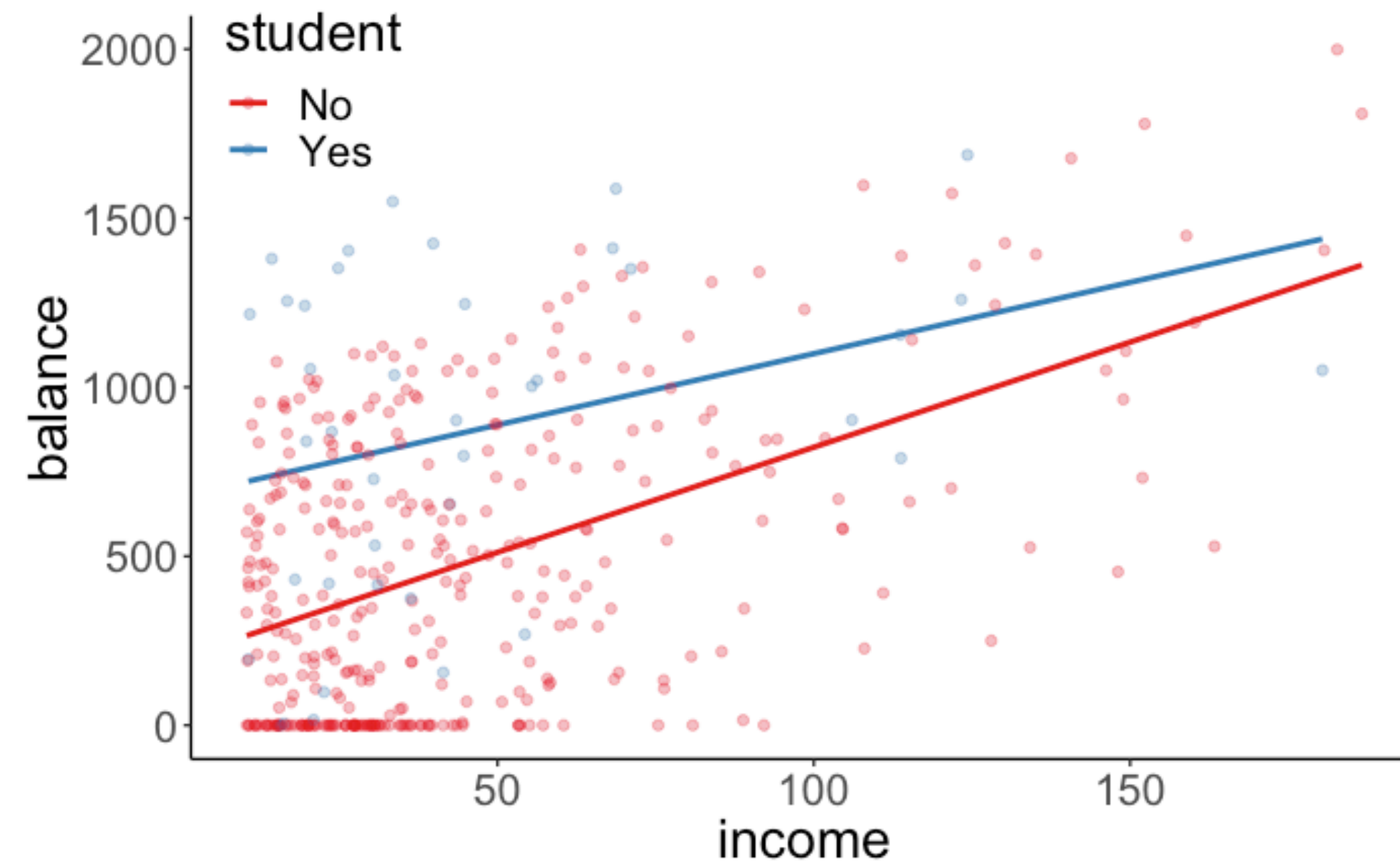
fit1 and fit2 are identical!

How to report results of interaction

There is no evidence that the relationship between income and balance differs for students versus non-students, $F(1, 396) = 1.33$, $p = 0.25$.

For *students*, each additional \$1000 of income is associated with a \$4.21 higher average credit card balance.

For *non-students*, each additional \$1000 income is associated with \$6.22 higher average credit card balance.



There was no evidence that student status *moderated* the relationship between income and balance, $F(1, 396) = 1.33$, $p = 0.25$.

lm () output

lm() output

```
1 lm(formula = balance ~ income + student + income:student, data = df.credit) %>%  
2 summary()
```

```
Call:  
lm(formula = balance ~ income + student + income:student,  
data = df.credit)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-773.39 -325.70  -41.13   321.65   814.04   
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept)   200.6232    33.6984   5.953 5.79e-09 ***  
income         6.2182     0.5921  10.502 < 2e-16 ***  
studentYes    476.6758   104.3512   4.568 6.59e-06 ***  
income:studentYes -1.9992    1.7313  -1.155  0.249      
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 391.6 on 396 degrees of freedom  
Multiple R-squared:  0.2799, Adjusted R-squared:  0.2744   
F-statistic: 51.3 on 3 and 396 DF,  p-value: < 2.2e-16
```



```
1 fit_c = lm(formula = balance ~ student + income:student, data = df.credit)  
2 fit_a = lm(formula = balance ~ income + student + income:student, data = df.credit)  
3  
4 anova(fit_c, fit_a)
```

```
1 fit_c = lm(formula = balance ~ income + student, data = df.credit)  
2 fit_a = lm(formula = balance ~ income + student + income:student, data = df.credit)  
3  
4 anova(fit_c, fit_a)
```

lm() output

```
1 lm(formula = balance ~ income + student + income:student, data = df.credit) %>%  
2 summary()
```

```
Call:  
lm(formula = balance ~ income + student + income:student,  
data = df.credit)  
  
Residuals:  
      Min       1Q   Median       3Q      Max   
-773.39 -325.70  -41.13   321.65   814.04   
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept)    200.6232     33.6984   5.953 5.79e-09 ***  
income           6.2182      0.5921  10.502 < 2e-16 ***  
studentYes      476.6758    104.3512   4.568 6.59e-06 ***  
income:studentYes -1.9992      1.7313  -1.155  0.249      
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 391.6 on 396 degrees of freedom  
Multiple R-squared:  0.2799, Adjusted R-squared:  0.2744   
F-statistic: 51.3 on 3 and 396 DF,  p-value: < 2.2e-16
```

```
1 fit_c = lm(formula = balance ~ 1, data = df.credit)  
2 fit_a = lm(formula = balance ~ income + student + income:student, data = df.credit)  
3  
4 anova(fit_c, fit_a)
```

Analysis of Variance Table

Model 1: balance ~ 1

Model 2: balance ~ 1 + income

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	399	84339912				
2	398	66208745	1	18131167	108.99	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

**deterministic mapping
between t and F**

$$t^2 = F$$

$$10.44^2 = 108.99$$

anova() gives me F s ?
but lm() gives me t s

```
Call:
lm(formula = balance ~ 1 + income, data = df.credit)

Residuals:
    Min       1Q   Median       3Q      Max
-803.64 -348.99 -54.42  331.75 1100.25

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  246.5148    33.1993   7.425  6.9e-13 ***
income         6.0484     0.5794  10.440 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 407.9 on 398 degrees of freedom
Multiple R-squared:  0.215,    Adjusted R-squared:  0.213
F-statistic: 109 on 1 and 398 DF,  p-value: < 2.2e-16
```


lm () output

```
1 lm(formula = balance ~ income + student + income:student, data = df.credit) %>%  
2 summary()
```

```
Call:  
lm(formula = balance ~ income + student + income:student, data =  
df.credit)  
  
Residuals:  
    Min       1Q   Median       3Q      Max   
-773.39 -325.70  -41.13   321.65   814.04  
  
Coefficients:  
            Estimate Std. Error t value Pr(>|t|)      
(Intercept)   200.6232    33.6984   5.953 5.79e-09 ***  
income         6.2182     0.5921  10.502 < 2e-16 ***  
studentYes    476.6758   104.3512   4.568 6.59e-06 ***  
income:studentYes -1.9992     1.7313  -1.155  0.249      
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 391.6 on 396 degrees of freedom  
Multiple R-squared:  0.2799,    Adjusted R-squared:  0.2744   
F-statistic: 51.3 on 3 and 396 DF,  p-value: < 2.2e-16
```

- runs many hypothesis tests at the same time
- increases the danger of making a type-I error (incorrectly rejecting the H_0)
- will not give us p-values for mixed effects models ...

The model comparison approach

- allows to formulate hypotheses as specific comparisons between candidate models
- is more flexible: we could test a model with 2 predictors vs. a model with 4 predictors (= 2 df difference)
- gives us insight into the underlying statistical procedure

lm () output

very important

```
Call:
lm(formula = balance ~ income + student + income:student,
    data = df.credit)

Residuals:
    Min       1Q   Median       3Q      Max
-773.39 -325.70  -41.13   321.65   814.04

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    200.6232    33.6984   5.953 5.79e-09 ***
income          6.2182     0.5921  10.502 < 2e-16 ***
studentYes     476.6758    104.3512   4.568 6.59e-06 ***
income:studentYes -1.9992     1.7313  -1.155  0.249
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 391.6 on 396 degrees of freedom
Multiple R-squared:  0.2799, Adjusted R-squared:  0.2744
F-statistic: 51.3 on 3 and 396 DF,  p-value: < 2.2e-16
```

what does this mean?

not the overall effect of income!

instead the predicted effect of income for non-students (student = 0)

we'll talk more about the difference between simple/conditional effects and main effects next time!

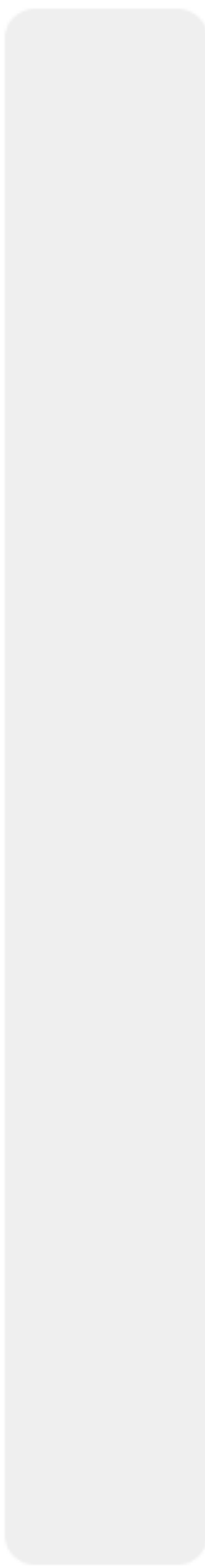
Summary

- Quick recap
- Multiple regression
 - two continuous predictors
 - one categorical predictor
 - one continuous and one categorical predictor
- Interactions
- `lm()` output

Feedback

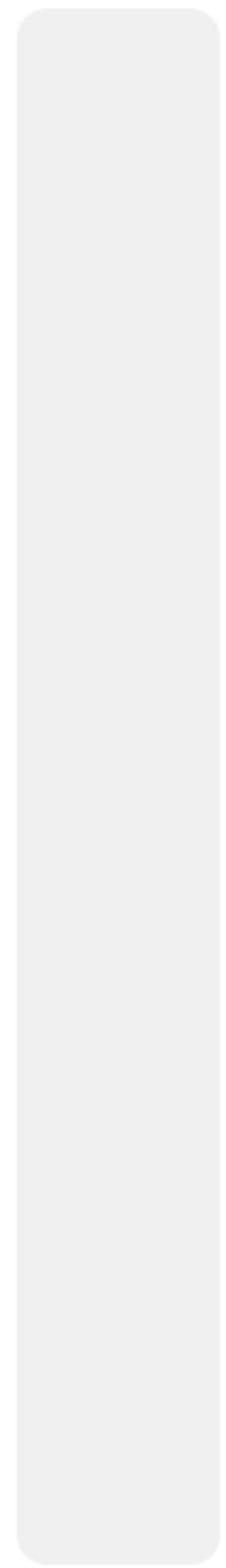


0%



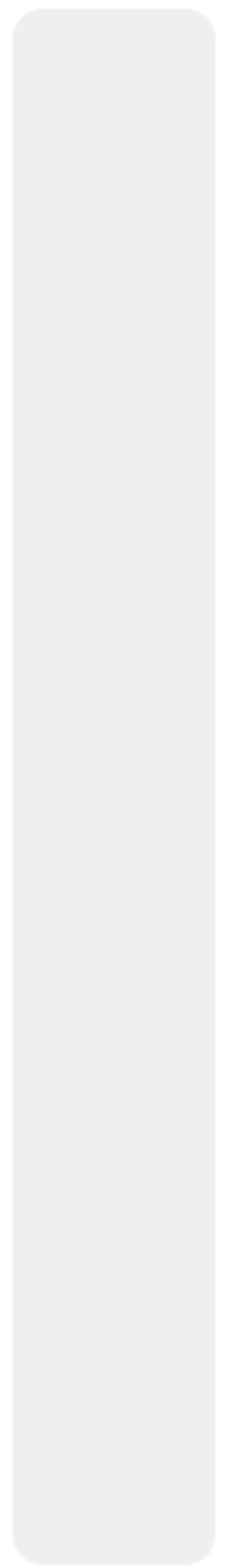
much too slow

0%



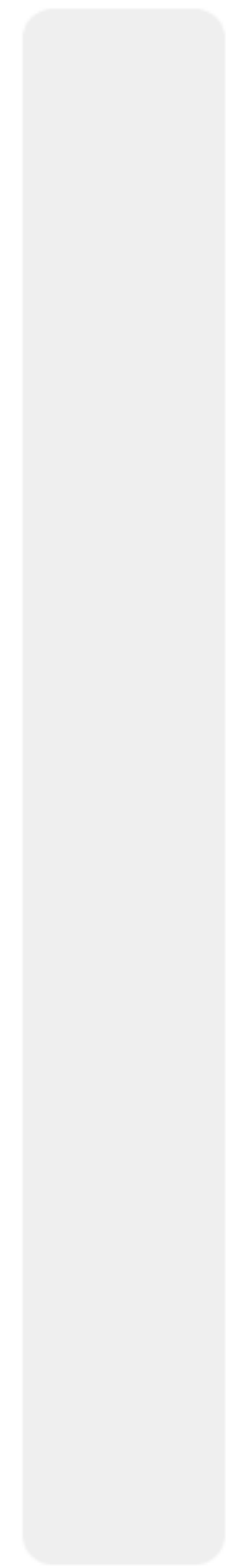
a little too slow

0%



just right

0%



a little too fast

0%



much too fast



Thank you!